

A PROFINITE PRESENTATION OF THE ABSOLUTE GALOIS GROUP OF $\mathbf{Q}_2$

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ABSTRACT. Let $\omega_2 \in \widehat{\mathbf{Z}}$ be the idempotent whose 2-adic coordinate is 1 and whose odd-primary coordinates are 0. With the conventions $x^g = g^{-1}xg$ and $[x, y] = x^{-1}y^{-1}xy$, define

$$\sigma_2 = \sigma^{\omega_2}, \quad g_0 = \sigma_2^2, \quad u_i = (x_i \tau)^{\omega_2}, \quad d_0 = u_0 x_0^{-1}, \quad z_0 = x_0^{\sigma_2}, \quad c_0 = [d_0, z_0],$$

$$d_g = d_0^{g_0}, \quad h_c = [d_g, d_0], \quad h_0 = x_0^{g_0} x_0 d_g d_0^2 h_c.$$

We prove that the absolute Galois group $G_{\mathbf{Q}_2}$ is the profinite group generated by σ, τ, x_0, x_1 , with the closed normal subgroup generated by x_0, x_1 pro-2, and with relations

$$\tau^\sigma = \tau^2, \quad h_0 u_1^{-1} x_1^\sigma c_0 = 1.$$

The displayed relation has also been computationally verified. The generator σ is normalized as geometric Frobenius; with $x^g = g^{-1}xg$ this is exactly the normalization in which $\tau^\sigma = \tau^2$.

The proof includes a marked dyadic Demushkin normalization for $G_{\mathbf{Q}_2}(2)$: standard generators a, s, y are constructed with $a^2 s^4 [s, y] = 1$ and $\nu_{\text{ur}}(a, s, y) = (-2, 1, 0)$. Section 3.1 derives this from Labute’s abstract classification and local reciprocity. The key structural step is the surjectivity of $\text{Aut}(D_0) \rightarrow \text{Aut}(D_0^{\text{ab}}, \chi_0)$ for the standard dyadic Demushkin group. It is proved uniformly from cyclotomic powering of the three peripheral classes of the thrice-punctured line, followed by a square-root and HNN lifting argument.

With the full marked pro-2 boundary fixed, evaluation of the exact relators in finite semidirect Heisenberg groups gives a natural three-term duality complex for every finite-dimensional $\mathbf{F}_2$ -module. A direct finite-word conjugation calculation gives the required Stokes identity; its correction terms cancel only after summing the tame and wild relators. The dyadic determinant obstruction is computed from the local square-class filtration. The involution-stabilized orbit cocycle and its normalized Shapiro–Evens comparison are made explicit. A constrained Fourier–Gauss formula converts affine quadratic phases into central covers of strictly smaller marked targets. Strong induction then gives equality of all finite surjection counts, and profinite reconstruction gives the isomorphism.

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1. INTRODUCTION AND MAIN THEOREM

For odd primes p , the absolute Galois group of $\mathbf{Q}_p$ admits an explicit four-generator presentation of Jannsen–Wingberg type; see [1, Chapter VII]. The dyadic case is exceptional because the odd-prime formula uses an exponent containing $(p+1)/2$. We give a separate dyadic relation.

Throughout, exponentiation by an element of $\widehat{\mathbf{Z}}$ has its usual profinite meaning. Let

$$\omega_2 = (1 \text{ in } \mathbf{Z}_2, 0 \text{ in } \mathbf{Z}_\ell \text{ for every odd prime } \ell).$$

Thus $\omega_2^2 = \omega_2$. For the finite verification modulus

$$M = 85667662080 = 2^8 \cdot 3^2 \cdot 5 \cdot 7 \cdot 11 \cdot 13 \cdot 17 \cdot 19 \cdot 23,$$

its residue is

$$\omega_2 \equiv 40491355905 \pmod{M}.$$

The integer is only a serialization of the profinite exponent; all arguments below use the genuine element $\omega_2 \in \widehat{\mathbf{Z}}$.

We use the conventions

$$x^g = g^{-1}xg, \quad [x, y] = x^{-1}y^{-1}xy.$$

The unramified coordinate is normalized geometrically throughout: if Φ_{ar} is arithmetic Frobenius and $\Phi_{\text{geom}} = \Phi_{\text{ar}}^{-1}$, then $\nu_{\text{ur}}(\Phi_{\text{geom}}) = 1$. With the right-conjugation convention above, a standard tame generator satisfies

$$\tau^{\Phi_{\text{geom}}} = \tau^q.$$

Thus the displayed relation $\tau^\sigma = \tau^2$ means that σ maps to geometric Frobenius. Put

$$\sigma_2 = \sigma^{\omega_2}, \quad u_i = (x_i \tau)^{\omega_2} \ (i = 0, 1), \quad d_0 = u_0 x_0^{-1}, \quad (1)$$

$$z_0 = x_0^{\sigma_2}, \quad c_0 = [d_0, z_0], \quad g_0 = \sigma_2^2, \quad d_g = d_0^{g_0}, \quad (2)$$

$$h_c = [d_g, d_0], \quad h_0 = x_0^{g_0} x_0 d_g d_0 d_0^2 h_c. \quad (3)$$

Proposition 1.1 (Marked dyadic Demushkin normalization). *Let $D = G_{\mathbf{Q}_2}(2)$ and let $\nu_{\text{ur}} : D \twoheadrightarrow \mathbf{Z}_2$ be normalized by sending geometric Frobenius to 1. There exist topological generators a, s, y such that*

$$D \cong \langle a, s, y \mid a^2 s^4 [s, y] = 1 \rangle_{\text{pro-2}}, \quad \nu_{\text{ur}}(a) = -2, \quad \nu_{\text{ur}}(s) = 1, \quad \nu_{\text{ur}}(y) = 0. \quad (4)$$

The proof is given in [section 3.1](#).

Theorem 1.2 (Presentation theorem). *The absolute Galois group $G_{\mathbf{Q}_2}$ is isomorphic to the profinite group topologically generated by*

$$\sigma, \tau, x_0, x_1,$$

for which the closed normal subgroup generated by x_0, x_1 is pro-2, and which is subject to the two relations

$$\tau^\sigma = \tau^2 \quad (5)$$

and

$$h_0 u_1^{-1} x_1^\sigma c_0 = 1, \quad (6)$$

where the auxiliary words are defined in (1)–(3).

Remark 1.3 (Role of the computational verification). *A separate computational verification of the displayed word exists but is not reproduced here. It is independent of the proof below and is not used as an input to the marked Demushkin normalization or to any mathematical inference.*

The condition that the normal closure of x_0, x_1 be pro-2 is part of the presentation. In a finite quotient it is equivalent, for an epimorphism, to requiring the images of x_0, x_1 to lie in the 2-core $O_2(G)$.

The proof proceeds through finite quotients. We first construct the candidate as an honest profinite group and choose compatible tame and maximal pro-2 markings. We then fix their common boundary and compare finite lifting problems through the marked wild 2-subgroup. Candidate lifts are controlled by a finite Fox–Heisenberg word complex, whereas local lifts are controlled by continuous Galois cohomology. The comparison is uniform in the target and is stable under exact-image subtraction, affine constraints, and auxiliary central covers. Organization. [Sections 2](#) and [3](#) construct the candidate, identify its tame quotient and maximal pro-2 quotient, and prove the full marked normalization [Proposition 1.1](#). [Section 4](#) fixes the common finite boundary used throughout the induction. The two source-specific lifting theories and their natural dualities are developed in [section 5](#); the dyadic quadratic obstruction is calculated in [section 6](#). The finite target is reduced to one elementary module layer and one Frattini layer, together with its invariant scalar central pushouts, in [section 7](#). Constrained Fourier inversion and the global exact-image induction occupy [sections 8](#) and [9](#). The final finite-quotient and reconstruction steps are in [section 10](#).

2. THE CANDIDATE AS A PROFINITE GROUP

2.1. The marked quotient construction. Let F be the free profinite group on σ, τ, x_0, x_1 . Call an epimorphism

$$\varphi : F \twoheadrightarrow G$$

to a finite group *admissible* if the distinguished tuple generates G , satisfies (5) and (6), and the normal closure of $\varphi(x_0), \varphi(x_1)$ is a 2-group. Let $\mathcal{Q}_A$ be the class of all admissible finite quotients and set

$$N_A = \bigcap_{\varphi \in \mathcal{Q}_A} \ker \varphi, \quad \Gamma_A = F/N_A. \quad (7)$$

Let W_A denote the closed normal subgroup of Γ_A generated by x_0, x_1 .

Lemma 2.1 (Finite subdirect closure). *If $F \twoheadrightarrow G_i$ are finitely many admissible quotients and Y is the image of F in $\prod_i G_i$, then $F \twoheadrightarrow Y$ is admissible and its kernel is the intersection of the given kernels.*

Proof. The two relations hold coordinatewise. The distinguished tuple generates Y . The normal closure of x_0, x_1 in Y lies in

$$Y \cap \prod_i O_2(G_i),$$

which is a finite 2-group. The assertion about kernels is immediate. $\square$

Lemma 2.2 (Cofinality). *Every finite quotient of Γ_A is an admissible marked quotient. In particular, W_A is pro-2.*

Proof. Let $K \triangleleft_{\text{open}} F$ contain N_A . For every $g \notin K$, the definition of N_A gives an admissible quotient whose kernel does not contain g . Compactness of $F \setminus K$ gives finitely many such kernels whose intersection is contained in K . By lemma 2.1, this intersection is itself the kernel of an admissible quotient. Therefore F/K is a quotient of an admissible quotient, and admissibility passes to quotients. The last assertion follows because every finite image of W_A is a 2-group. $\square$

Proposition 2.3 (Epimorphic finite-quotient semantics). *For every finite group G , epimorphisms $\Gamma_A \twoheadrightarrow G$ are in natural bijection with generating quadruples $(\sigma, \tau, x_0, x_1) \in G^4$ satisfying (5), (6), and $x_0, x_1 \in O_2(G)$.*

Proof. An epimorphism sends W_A to a normal 2-subgroup of G , hence into $O_2(G)$. Conversely, a generating quadruple with $x_0, x_1 \in O_2(G)$ has normal 2-group closure and defines an admissible quotient in $\mathcal{Q}_A$. $\square$

Remark 2.4. *The corresponding statement for arbitrary, non-surjective homomorphisms into a fixed ambient group is false: the image of W_A is normal in the image subgroup, but need not lie in the 2-core of the ambient group. This is why the proof below keeps track of exact images and their induced markings.*

2.2. A reconstruction lemma.

Lemma 2.5 (One-sided profinite reconstruction). *Let P be a topologically finitely generated profinite group and let Q be any profinite group. If*

$$|\text{Sur}(P, H)| = |\text{Sur}(Q, H)|$$

for every finite group H , then $P \cong Q$.

Proof. Since P is finitely generated, $\text{Sur}(P, H)$ and hence $\text{Sur}(Q, H)$ are finite for every finite H . The two groups have exactly the same finite quotients.

For $n \geq 1$, let $R_n(P)$ be the intersection of the kernels of all maps from P to groups of order at most n , and put $P_n = P/R_n(P)$. The P_n form a cofinal inverse system of finite quotients of P . Every P_n is a quotient of Q . For any finite set of indices, a sufficiently large P_m maps onto all corresponding P_n ; composing an epimorphism $Q \twoheadrightarrow P_m$ gives a compatible family on that finite set. Compactness of the inverse limit of the finite nonempty sets $\text{Sur}(Q, P_n)$ yields a compatible family and therefore an epimorphism $Q \twoheadrightarrow P$.

Apply the same argument to the directed system of all finite quotients of Q . The sets of epimorphisms from P to those quotients are finite and have the finite intersection property, so there is an epimorphism $P \twoheadrightarrow Q$. The composite $P \twoheadrightarrow Q \twoheadrightarrow P$ is a surjective endomorphism of a finitely generated profinite group. Such groups are Hopfian; see, for example, [4, Section 2.5]. Hence the composite is an isomorphism, and so is $P \twoheadrightarrow Q$. $\square$

3. THE TAME AND MAXIMAL PRO-2 QUOTIENTS

Let

$$T_{\text{tame}} = \langle \sigma, \tau \mid \tau^\sigma = \tau^2 \rangle_{\text{prof}}$$

denote the finite-quotient tame group.

Lemma 3.1 (Finite tame quotients). *In every finite quotient generated by s, t with $t^s = t^2$, the element t has odd order, and the quotient has the form*

$$C_e \rtimes C_n, \quad e \text{ odd}, \quad s^{-1}ts = t^2.$$

Every normal 2-subgroup of such a quotient is central and lies in the unramified cyclic direction.

Proof. Conjugation preserves order, whereas $|t^2| = |t|/\gcd(2, |t|)$, so $|t|$ is odd. Let $I = \langle t \rangle$ and suppose sI has order n . Then $s^n = t^m$. Since s^n commutes with s , the relation $t^s = t^2$ gives $t^m = t^{2^m}$, hence $m \equiv 0 \pmod{|t|}$. Thus the extension splits.

If N is a normal 2-subgroup, then $N \cap I = 1$ and $[N, I] \leq N \cap I = 1$. Hence N centralizes I . It is therefore contained in the 2-primary part of the kernel of the Frobenius action on I , which is central and unramified. $\square$

Proposition 3.2 (Common tame quotient). *There are canonical isomorphisms*

$$\Gamma_A/W_A \cong T_{\text{tame}} \cong G_{\mathbf{Q}_2}/W_F,$$

where W_F is wild inertia.

Proof. In the quotient by W_A , the elements x_0, x_1 vanish. By lemma 3.1, τ has odd order in every finite quotient, so $\tau_2^\omega = 1$. Hence $u_i = d_0 = z_0 = c_0 = d_g = h_c = h_0 = 1$; the words σ_2 and g_0 may remain nontrivial, but they only conjugate the identity. The wild relation therefore disappears. The local statement is the standard description of the tame quotient in the geometric normalization: tame inertia is pro-prime-to-2 cyclic, and with the convention $x^g = g^{-1}xg$ geometric Frobenius acts by squaring. $\square$

Lemma 3.3 (The wild subgroup is the characteristic 2-core). *The tame group T_{tame} has no nontrivial closed normal pro-2 subgroup. Consequently*

$$O_2(\Gamma_A) = W_A, \quad O_2(G_{\mathbf{Q}_2}) = W_F.$$

In particular, the marked wild subgroups of the two sources are characteristic.

Proof. Let $I = \overline{\langle \tau \rangle} \triangleleft T_{\text{tame}}$ be tame inertia; it is pro-odd. If $N \triangleleft T_{\text{tame}}$ is pro-2, then

$$N \cap I = 1, \quad [N, I] \leq N \cap I = 1,$$

so N centralizes I . Its image in $T_{\text{tame}}/I \cong \widehat{\mathbf{Z}}$ lies in the 2-primary factor and has a 2-adic geometric Frobenius coordinate $a \in \mathbf{Z}_2$. For every $m \geq 1$, use the finite inertia quotient of order

$$e_m = 2^{2^m} - 1.$$

Centrality forces $2^a \equiv 1 \pmod{e_m}$. The order of 2 modulo e_m is exactly 2^m , and hence $a \in 2^m \mathbf{Z}_2$ for every m . Thus $a = 0$, so $N \leq I$, and therefore $N = 1$.

The subgroup W_A is normal pro-2 and $\Gamma_A/W_A \cong T_{\text{tame}}$. Every normal pro-2 subgroup of Γ_A maps trivially to T_{tame} and is contained in W_A . This proves $O_2(\Gamma_A) = W_A$. The identical argument applied to $1 \rightarrow W_F \rightarrow G_{\mathbf{Q}_2} \rightarrow T_{\text{tame}} \rightarrow 1$ gives the local assertion. $\square$

3.1. The marked dyadic Demushkin normalization. We now prove [Proposition 1.1](#). The argument separates the intrinsic Demushkin orientation from the chosen integral marking. The structural input is a uniform lifting theorem for orientation-preserving automorphisms of the abelianization of the standard rank-three dyadic Demushkin group.

Put

$$D_0 = \langle A, S, Y \mid A^2 S^4 [S, Y] = 1 \rangle_{\text{pro-2}} \quad (8)$$

and

$$B = D_0^{\text{ab}} = \langle \bar{A}, \bar{S}, \bar{Y} \mid 2\bar{A} + 4\bar{S} = 0 \rangle_{\mathbf{Z}_2}. \quad (9)$$

Let

$$t = \bar{A} + 2\bar{S}, \quad \eta = (-3)^{-1} \in \mathbf{Z}_2^\times. \quad (10)$$

Then

$$B = C_2 t \oplus \mathbf{Z}_2 \bar{S} \oplus \mathbf{Z}_2 \bar{Y}. \quad (11)$$

Lemma 3.4 (Standard dyadic group and its orientation). *The group D_0 is the standard rank-three, $q = 2$ Demushkin group, and $G_{\mathbf{Q}_2}(2)$ is abstractly isomorphic to D_0 . The canonical Demushkin orientation of D_0 factors through B and is the character*

$$\chi_0(t) = -1, \quad \chi_0(\bar{S}) = 1, \quad \chi_0(\bar{Y}) = \eta. \quad (12)$$

Equivalently,

$$\chi_0(\bar{A}) = -1, \quad \chi_0(\bar{S}) = 1, \quad \chi_0(\bar{Y}) = (-3)^{-1}.$$

Proof. The presentation (8) is the odd-rank dyadic case of Labute's classification, with parameter $q = 2$ and secondary exponent 4; see [2, Theorem 8]. The canonical character for the odd-rank normal form is computed in [2, Theorem 4, case (2)]: for the relation $x_1^2 x_2^{2^f} [x_2, x_3]$ it takes the values $-1, 1, (1 - 2^f)^{-1}$ on x_1, x_2, x_3 . With $f = 2$ and the standing commutator convention, this gives $-1, 1, (1 - 4)^{-1} = (-3)^{-1}$ on A, S, Y , respectively. The maximal pro-2 Galois group of $\mathbf{Q}_2$ is the rank-three local Demushkin group with these parameters, so the abstract isomorphism follows from the same classification. $\square$

Lemma 3.5 (Marked abelianization, orientation, and initial form). *Use arithmetic local reciprocity rec , normalized so that $\text{rec}(2)$ is arithmetic Frobenius, and normalize ν_{ur} geometrically. Put*

$$\bar{s} = \text{rec}(2)^{-1} = \text{rec}(1/2), \quad \bar{e} = \text{rec}(-1), \quad \bar{y} = \text{rec}(-3), \quad \bar{a} = \bar{e}\bar{s}^{-2} = \text{rec}(-4).$$

Then the pro-2 abelianization of $D = G_{\mathbf{Q}_2}(2)$ is identified with B by

$$\bar{A} \mapsto \bar{a}, \quad \bar{S} \mapsto \bar{s}, \quad \bar{Y} \mapsto \bar{y}.$$

Under this identification,

$$\nu_{\text{ur}}(\bar{a}, \bar{s}, \bar{y}) = (-2, 1, 0), \quad \chi_D(\bar{a}, \bar{s}, \bar{y}) = (-1, 1, (-3)^{-1}), \quad (13)$$

where χ_D is the canonical Demushkin orientation. In the basis of $H^1(D, \mathbf{F}_2)$ dual to $(\bar{a}, \bar{s}, \bar{y})$, the Demushkin quadratic initial form is

$$\alpha^2 + \beta\gamma + \gamma\beta.$$

Consequently a Demushkin relator for lifts of these classes has the same quadratic initial form as

$$r_0 = A^2 S^4 [S, Y],$$

namely $A^2 + [S, Y]$ in degree two. Moreover the pair

$$(\nu_{\text{ur}}, \chi_D) : B \longrightarrow \mathbf{Z}_2 \times \mathbf{Z}_2^\times$$

is injective.

Proof. Since ν_{ur} is normalized by geometric Frobenius while $\text{rec}(2)$ is arithmetic Frobenius, one has $\nu_{\text{ur}}(\text{rec}(2)) = -1$. Hence

$$\nu_{\text{ur}}(\bar{s}) = 1, \quad \nu_{\text{ur}}(\bar{y}) = 0, \quad \nu_{\text{ur}}(\bar{a}) = -2.$$

The pro-2 completion of $\mathbf{Q}_2^\times$ is

$$\widehat{\mathbf{Q}_2^\times}^{(2)} \cong \langle 2 \rangle_{\mathbf{Z}_2} \oplus \langle -1 \rangle_{C_2} \oplus \langle -3 \rangle_{\mathbf{Z}_2}.$$

The single relation among $\bar{A}, \bar{S}, \bar{Y}$ is

$$\bar{a}^2 \bar{s}^4 = \text{rec}((-4)^2 (1/2)^4) = \text{rec}(1) = 1,$$

that is, additively, $2\bar{A} + 4\bar{S} = 0$.

For a local Demushkin group the canonical orientation is the 2-adic cyclotomic character. With the reciprocity convention above, local reciprocity gives

$$\chi_D(\text{rec}(u)) = u^{-1} \quad (u \in \mathbf{Z}_2^\times),$$

while the uniformizer component acts trivially on the totally ramified extension $\mathbf{Q}_2(\mu_{2^\infty})/\mathbf{Q}_2$. This proves the orientation row in (13).

Modulo squares, the classes are

$$\bar{a} \equiv [-1], \quad \bar{s} \equiv [2], \quad \bar{y} \equiv [-3].$$

For odd units u, v , write

$$\epsilon(u) = \frac{u-1}{2} \bmod 2, \quad \omega(u) = \frac{u^2-1}{8} \bmod 2.$$

The dyadic Hilbert-symbol formula

$$(2^\alpha u, 2^\beta v)_2 = (-1)^{\epsilon(u)\epsilon(v) + \alpha\omega(v) + \beta\omega(u)}$$

gives, in the square-class basis $(-1, 2, -3)$,

$$(-1, -1)_2 = -1, \quad (2, -3)_2 = -1,$$

with all other pairings among these basis elements trivial, up to symmetry. Under Kummer theory, cup product is the Hilbert symbol. Thus the dual basis has Bockstein/cup initial form $\alpha^2 + \beta\gamma + \gamma\beta$, which is exactly the degree-two initial form of $A^2 S^4[S, Y]$.

Finally, in the decomposition (11),

$$\nu_{\text{ur}}(t, \bar{S}, \bar{Y}) = (0, 1, 0), \quad \chi_D(t, \bar{S}, \bar{Y}) = (-1, 1, \eta).$$

The element η topologically generates $1 + 4\mathbf{Z}_2$, because $v_2(\eta - 1) = 2$. The two displayed characters therefore detect the three direct summands of (11), proving injectivity. $\square$

Lemma 3.6 (Cyclotomic powering of the three peripheral classes). *Let*

$$\Delta = \langle P, T \rangle_{\text{pro-2}}, \quad C = (PT)^{-1}.$$

For every $u \in \mathbf{Z}_2^\times$ there are an automorphism $\varphi_u \in \text{Aut}(\Delta)$ and elements $c_P, c_T, c_C \in \Delta$ such that

$$\varphi_u(P) = c_P^{-1} P^u c_P, \quad \varphi_u(T) = c_T^{-1} T^u c_T, \quad \varphi_u(C) = c_C^{-1} C^u c_C. \quad (14)$$

Proof. Identify Δ with the maximal pro-2 geometric fundamental group of

$$X = \mathbf{P}_{\mathbf{Q}}^1 \setminus \{0, 1, \infty\},$$

with P, T, C the three compatible peripheral inertia generators. The Galois action on the geometric fundamental group gives

$$G_{\mathbf{Q}} \longrightarrow \text{Out}(\Delta).$$

The 2-adic cyclotomic character is surjective, since

$$\text{Gal}(\mathbf{Q}(\mu_{2^\infty})/\mathbf{Q}) \cong \mathbf{Z}_2^\times.$$

Each of the three cusps is rational, so its peripheral conjugacy class is preserved. The corresponding decomposition group acts on the procyclic inertia group through the cyclotomic character; see [8, Section 3.3 and Definition 37]. Concretely, if z is a local parameter and a chosen inertia generator sends $z^{1/2^n}$ to $\zeta_{2^n} z^{1/2^n}$, then the arithmetic action of g replaces ζ_{2^n} by $\zeta_{2^n}^{\chi_2(g)}$. With the opposite convention for representing the outer action, the exponent is $\chi_2(g)^{-1}$. Accordingly choose g with $\chi_2(g) = u$ in the first convention and with $\chi_2(g) = u^{-1}$ in the second. In either convention the resulting outer automorphism sends each peripheral generator to a conjugate of its u -th power. Choosing a representative of the outer class gives (14). $\square$

Lemma 3.7 (Square-root and HNN lifting). *For every $u \in \mathbf{Z}_2^\times$ there is an automorphism $\Psi_u \in \text{Aut}(D_0)$ such that, on B ,*

$$\bar{A} \mapsto u\bar{A}, \quad \bar{S} \mapsto u\bar{S}. \quad (15)$$

Proof. Start with φ_u and the conjugators from lemma 3.6. Consider

$$E_\square = \langle P, T, A \mid PTA^2 = 1 \rangle_{\text{pro-2}} \cong \langle P, A \rangle_{\text{pro-2}}, \quad C = A^2.$$

View the words c_P, c_T, c_C in $E_\square$ via P, T , and define

$$P \mapsto \varphi_u(P), \quad T \mapsto \varphi_u(T), \quad A \mapsto c_C^{-1} A^u c_C.$$

The defining relation is preserved, because

$$(c_C^{-1} A^u c_C)^2 = c_C^{-1} C^u c_C = \varphi_u(C) = (\varphi_u(P) \varphi_u(T))^{-1}.$$

Modulo the Frattini subgroup, the free basis P, A is sent to $u\bar{P}, u\bar{A}$, hence to $\bar{P}, \bar{A}$. The pro-2 Burnside basis theorem and Hopficity therefore show that this endomorphism is an automorphism; call it θ_u .

In the free pro-2 group $E = \langle S, A \rangle_{\text{pro-2}}$ put

$$P = S^3, \quad T = S^{-3} A^{-2}.$$

Since $3 \in \mathbf{Z}_2^\times$, the pair P, A is again a free basis of E , and $PTA^2 = 1$. For every odd integer m , the power map $x \mapsto x^m$ is bijective on a pro-2 group, with inverse $x \mapsto x^{m^{-1}}$ for $m^{-1} \in \mathbf{Z}_2^\times$. Comparing the unique cube roots of the two sides of (14) therefore gives

$$\theta_u(S) = c_P^{-1} S^u c_P, \quad \theta_u(T) = c_T^{-1} T^u c_T.$$

The relation of D_0 is equivalently

$$Y^{-1}SY = T, \quad T = S^{-3}A^{-2}. \quad (16)$$

Define an endomorphism of the presentation (16) by

$$\Psi_u|_E = \theta_u, \quad \Psi_u(Y) = c_P^{-1} Y c_T.$$

Then

$$\Psi_u(Y)^{-1} \Psi_u(S) \Psi_u(Y) = c_T^{-1} Y^{-1} S^u Y c_T = c_T^{-1} T^u c_T = \Psi_u(T),$$

so the relation is respected. On the Frattini quotient of D_0 , the images of A, S, Y form a triangular basis with diagonal entries 1, 1, 1. Consequently Ψ_u is surjective and hence an automorphism. Its action on $\bar{A}, \bar{S}$ is (15). $\square$

Proposition 3.8 (Lifting the orientation-preserving abelian automorphisms). *The natural map*

$$\text{Aut}(D_0) \longrightarrow \text{Aut}(B, \chi_0) \quad (17)$$

is surjective. More explicitly, every element of $\text{Aut}(B, \chi_0)$ has the unique form

$$\alpha_{u,b}: \quad t \mapsto t, \quad \bar{S} \mapsto u\bar{S}, \quad \bar{Y} \mapsto \bar{Y} + b\bar{S}, \quad (18)$$

with $u \in \mathbf{Z}_2^\times$ and $b \in \mathbf{Z}_2$, and every $\alpha_{u,b}$ lifts to $\text{Aut}(D_0)$.

Proof. Because $v_2(\eta - 1) = 2$, the element η topologically generates $1 + 4\mathbf{Z}_2$. From (11) and (12) one obtains

$$\ker \chi_0 = \mathbf{Z}_2 \bar{S}.$$

The torsion subgroup of B is the order-two group generated by t . Hence an orientation-preserving automorphism fixes t , sends $\bar{S}$ to $u\bar{S}$ for a unique unit u , and sends $\bar{Y}$ to $\bar{Y} + b\bar{S}$ for a unique b . Conversely, every map (18) is an orientation-preserving automorphism of B .

The automorphism Ψ_u of lemma 3.7 sends $\bar{A}, \bar{S}$ to $u\bar{A}, u\bar{S}$, so it fixes $t = \bar{A} + 2\bar{S}$ and sends $\bar{S}$ to $u\bar{S}$. The canonical orientation is the action on the dualizing module and is therefore preserved by every automorphism of D_0 . Thus Ψ_u induces $\alpha_{u, b(u)}$ for some $b(u) \in \mathbf{Z}_2$.

For $b \in \mathbf{Z}_2$, define

$$\Theta_b(A) = A^{S^b}, \quad \Theta_b(S) = S, \quad \Theta_b(Y) = YS^b, \quad (19)$$

Here $A^{S^b} = (S^b)^{-1}AS^b$ is conjugation, not the product AS^b . This is an automorphism of the free pro-2 group on A, S, Y , with inverse Θ_{-b} . Moreover

$$[S, YS^b] = [S, Y]^{S^b},$$

and therefore

$$\Theta_b(A^2S^4[S, Y]) = (A^2)^{S^b}S^4[S, Y]^{S^b} = (A^2S^4[S, Y])^{S^b}.$$

It therefore descends to an automorphism of D_0 , inducing $\alpha_{1, b}$. The composite $\Theta_{b-b(u)} \circ \Psi_u$ induces $\alpha_{u, b}$, proving surjectivity. $\square$

Proof of proposition 1.1. By lemma 3.4, choose an abstract isomorphism

$$f : D \xrightarrow{\sim} D_0.$$

Use lemma 3.5 to identify D^{ab} with B , and use (9) for D_0^{ab} . Canonical orientations are functorial under group isomorphisms. The orientation row (13) and (12) therefore imply

$$\chi_0 \circ f_{\text{ab}} = \chi_D = \chi_0.$$

Thus $f_{\text{ab}} \in \text{Aut}(B, \chi_0)$. By proposition 3.8, choose $\gamma \in \text{Aut}(D_0)$ with

$$\gamma_{\text{ab}} = f_{\text{ab}}^{-1}.$$

Then

$$\beta = \gamma \circ f : D \xrightarrow{\sim} D_0$$

induces the identity on the marked abelianization B .

Set

$$a = \beta^{-1}(A), \quad s = \beta^{-1}(S), \quad y = \beta^{-1}(Y).$$

These elements topologically generate D , satisfy $a^2s^4[s, y] = 1$, and have the prescribed abelianized classes. Hence

$$\nu_{\text{ur}}(a, s, y) = (-2, 1, 0),$$

which proves proposition 1.1. $\square$

Remark 3.9 (Why the orientation cannot be omitted). *An un-oriented marked-relator normalization with only the abelianized relation and quadratic initial form is false. Let $F = \langle A, S, Y \rangle_{\text{pro-2}}$ and define*

$$\psi(A) = AY^{-4}, \quad \psi(S) = SY^2, \quad \psi(Y) = Y.$$

This is an automorphism, with inverse $A \mapsto AY^4, S \mapsto SY^{-2}, Y \mapsto Y$. For $r_0 = A^2S^4[S, Y]$, put $r = \psi^{-1}(r_0)$. On the free abelianization,

$$\psi_{\text{ab}}(2\bar{A} + 4\bar{S}) = 2(\bar{A} - 4\bar{Y}) + 4(\bar{S} + 2\bar{Y}) = 2\bar{A} + 4\bar{S}.$$

Moreover

$$r = (AY^4)^2(SY^{-2})^4[SY^{-2}, Y],$$

so, modulo the third Zassenhaus term, its initial form is $A^2 + [S, Y]$. Thus r has the same abelianized relation and quadratic initial form as r_0 , and its quotient is abstractly D_0 . However, under the isomorphism induced by ψ , the transported canonical orientation takes S to $\eta^2 \neq 1$. Therefore no isomorphism from this quotient to D_0 which is the identity on B can exist. The orientation-preserving lifting statement [proposition 3.8](#), rather than the false un-oriented normalization, is the exact structural input needed above.

Proposition 3.10 (Maximal pro-2 quotient). *The maximal pro-2 quotient of Γ_A is*

$$\Pi = \left\langle \sigma, x_0, x_1 \mid x_0^{\sigma^2} x_0 [x_1, \sigma] = 1 \right\rangle_{\text{pro-2}}. \quad (20)$$

The group Π is abstractly isomorphic to $G_{\mathbf{Q}_2}(2)$. Equipped with

$$\nu_2 : \Pi \rightarrow \mathbf{Z}_2, \quad \nu_2(\sigma) = 1, \quad \nu_2(x_0) = \nu_2(x_1) = 0, \quad (21)$$

it is isomorphic to the fully unramified marked pair $(G_{\mathbf{Q}_2}(2), \nu_{\text{ur}})$.

Proof. In a finite 2-group quotient, [lemma 3.1](#) forces $\tau = 1$, and ω_2 acts as the identity. Hence

$$u_i = x_i, \quad d_0 = 1, \quad c_0 = 1, \quad g_0 = \sigma^2, \quad d_g = h_c = 1,$$

and

$$h_0 = x_0^{\sigma^2} x_0.$$

Moreover

$$u_1^{-1} x_1^{\sigma} = x_1^{-1} x_1^{\sigma} = [x_1, \sigma].$$

Thus (6) becomes precisely the relator in (20). Conversely, every finite quotient of (20) is admissible after setting $\tau = 1$. The fully marked local identification is given by [corollary 3.12](#). $\square$

Proposition 3.11 (Nielsen transform to the pro-2 boundary). *For any pro-2 group with topological generators a, s, y and unique defining relation*

$$a^2 s^4 [s, y] = 1, \quad (22)$$

the Nielsen change

$$\sigma = s, \quad x_1 = y, \quad x_0 = s^{-2} a^{-1} \quad (23)$$

changes the closed normal closure of the relator to that of

$$x_0^{\sigma^2} x_0 [x_1, \sigma]. \quad (24)$$

Consequently [proposition 1.1](#) identifies $D = G_{\mathbf{Q}_2}(2)$ abstractly with Π .

Proof. Under (23), one has $a = x_0^{-1} \sigma^{-2}$ and therefore

$$a^2 s^4 = x_0^{-1} (x_0^{-1})^{\sigma^2}.$$

Writing $P = x_0^{\sigma^2} x_0$ and $Q = [x_1, \sigma]$, the inverse of the standard relator is QP . The words QP and PQ are cyclic conjugates, so they have the same closed normal closure. This proves the assertion. $\square$

Corollary 3.12 (Fully marked form). *The generators furnished by [proposition 1.1](#) and transformed by (23) satisfy*

$$\nu_{\text{ur}}(\sigma) = 1, \quad \nu_{\text{ur}}(x_0) = \nu_{\text{ur}}(x_1) = 0,$$

and (24) is the unique defining relation. Hence $(D, \nu_{\text{ur}}) \cong (\Pi, \nu_2)$.

Proof. The relation is [proposition 3.11](#). The values follow directly from (4):

$$\nu_{\text{ur}}(x_0) = -2\nu_{\text{ur}}(s) - \nu_{\text{ur}}(a) = 0.$$

$\square$

Remark 3.13 (The integral marking is not detected by the mod-2 cup form). *The more familiar unmarked Labute relator*

$$x_6^2 x_1^4 [x_1, \sigma]$$

has the same quadratic initial form $b^2 + ac + ca$ as (24). It does not, however, admit the marking

$$\nu(\sigma) = 1, \quad \nu(x_0) = \nu(x_1) = 0$$

required by the tame boundary. In those Labute variables the canonical orientation has values

$$\chi(x_0) = -1, \quad \chi(x_1) = 1, \quad \chi(\sigma) = (-3)^{-1}.$$

The abelianization is $\mathbf{Z}_2^2 \oplus C_2$, and $\bar{x}_1$ has infinite order. The asserted marking would nevertheless put x_1 in $\ker \nu \cap \ker \chi$, contradicting the injectivity of (ν_{ur}, χ) on D^{ab} . Thus agreement of the maximal pro-2 quotient as an unmarked Demushkin group is insufficient. The Nielsen transformation (23) replaces that relator by (24). It has the same mod-2 two-jet; its full unramified marking follows from Proposition 1.1.

Proposition 3.14 (Fully marked tame and pro-2 quotients). *Let*

$$\nu_t : T_{\text{tame}} \rightarrow \mathbf{Z}_2, \quad \nu_t(\sigma) = 1, \quad \nu_t(\tau) = 0,$$

and equip Π with (21). For each $\Gamma \in \{\Gamma_A, G_{\mathbf{Q}_2}\}$, the tame quotient map and the maximal pro-2 quotient map may be chosen so that their composites with ν_t and ν_2 are the same natural unramified character

$$\nu_\Gamma : \Gamma \rightarrow \mathbf{Z}_2.$$

The compatibility therefore holds after passage to every finite quotient C_{2^m} , not only modulo 2.

Proof. For the candidate, the assignment

$$\sigma \mapsto 1, \quad \tau, x_0, x_1 \mapsto 0$$

respects both defining relations and induces the displayed maps on the tame and pro-2 quotients. For the local source, use corollary 3.12. The tame geometric Frobenius and the pro-2 generator σ have the same full unramified image, while x_0, x_1 lie in its kernel. $\square$

Remark 3.15 (Why the full marking is load-bearing). *The construction*

$$\partial_{\text{bd}} = T_{\text{tame}} \times_{\mathbf{Z}_2} \Pi$$

uses the full marked isomorphism $(\Pi, \nu_2) \cong (G_{\mathbf{Q}_2}(2), \nu_{\text{ur}})$. An unmarked Demushkin classification identifies only the abstract pro-2 group; it does not fix the terminal comparison after passage to all finite quotients C_{2^m} of the unramified coordinate.

Let (a, b, c) be the basis of $H^1(\Pi, \mathbf{F}_2)$ dual to (σ, x_0, x_1) . The quadratic initial form of the relator in (20) is

$$b^2 + ac + ca.$$

Consequently the scalar cup/Bockstein table is

$$\mathcal{H}((a, b, c), (a', b', c')) = ac' + ca' + bb'. \quad (25)$$

This is the local Hilbert pairing on $\mathbf{Q}_2^\times / \mathbf{Q}_2^{\times 2}$ under the corresponding basis.

4. A COMMON BOUNDARY AND THE GLOBAL INDUCTION

The tame and pro-2 quotient maps cannot be fixed independently: they overlap in the entire unramified pro-2 quotient, not merely in its quadratic reduction. We encode that overlap once and for all by putting

$$\partial_{\text{bd}} = T_{\text{tame}} \times_{\mathbf{Z}_2} \Pi, \quad (26)$$

where the two maps to $\mathbf{Z}_2$ are those of [proposition 3.14](#). For $\Gamma \in \{\Gamma_A, G_{\mathbf{Q}_2}\}$, its tame and maximal pro-2 maps combine to an epimorphism

$$b_\Gamma : \Gamma \twoheadrightarrow \partial_{\text{bd}}. \quad (27)$$

Surjectivity follows from profinite Goursat. Any common quotient of T_{tame} and the pro-2 group Π is a pro-2 quotient of T_{tame} , hence a quotient of its maximal pro-2 quotient $\mathbf{Z}_2$; [proposition 3.14](#) shows that the full $\mathbf{Z}_2$ quotient occurs.

Fix a finite tame quotient H , an epimorphism $\alpha : T_{\text{tame}} \twoheadrightarrow H$, an elementary abelian 2-group E , and a homomorphism $\psi : \Pi \rightarrow E$. They define

$$\beta : \partial_{\text{bd}} \longrightarrow H \times E, \quad \beta(t, p) = (\alpha(t), \bar{\psi}(p)). \quad (28)$$

Definition 4.1 (Boundary-framed marked target). *A boundary-framed marked target is a tuple*

$$\mathcal{Y} = (Y, L_Y, \pi_Y, \theta_Y)$$

where $L_Y \triangleleft Y$ is a finite 2-group, $\pi_Y : Y \twoheadrightarrow H$ has kernel L_Y , and $\theta_Y : Y \rightarrow E$ is a homomorphism. Put

$$q_Y = (\pi_Y, \theta_Y) : Y \longrightarrow H \times E.$$

For $\Gamma \in \{\Gamma_A, G_{\mathbf{Q}_2}\}$ define

$$e_\Gamma^\beta(\mathcal{Y}) = \#\{f : \Gamma \twoheadrightarrow Y : q_Y f = \beta b_\Gamma\}. \quad (29)$$

Because the wild subgroup is the kernel of the tame quotient, (29) automatically sends W_A or W_F into L_Y . It also fixes the scalar data on the two different source groups without pretending that a single homomorphism has both groups as its domain.

If $J \leq Y$ projects onto H , then

$$\mathcal{J} = (J, J \cap L_Y, \pi_Y|_J, \theta_Y|_J)$$

is another boundary-framed marked target. Exact-image strata are therefore ordinary objects of the same category.

Theorem 4.2 (Boundary-framed exact-image theorem). *For every boundary-framed marked target $\mathcal{Y}$,*

$$e_{\Gamma_A}^\beta(\mathcal{Y}) = e_{G_{\mathbf{Q}_2}}^\beta(\mathcal{Y}).$$

The same equality holds for every exact-image target $\mathcal{J}$ defined above.

The proof is a global induction over all boundary-framed marked targets, ordered by $|L_Y|$. Auxiliary central covers need not lie inside the original ambient group, which is why the induction is stated globally.

5. TWO SOURCE-SPECIFIC LIFTING THEORIES

Let C be a finite lower target, let $\rho : \Gamma \twoheadrightarrow C$ satisfy the boundary equation, and let A be a finite elementary $\mathbf{F}_2[C]$ -module. Here *elementary* means that the additive group of A is a finite-dimensional $\mathbf{F}_2$ -vector space; no semisimplicity assumption is made. The two source groups have different cochain theories, and we keep them separate.

For $\Gamma = \Gamma_A$, varying the four distinguished generators through an extension by A gives the finite word complex

$$C_{\rho, A}^\bullet : 0 \longrightarrow A \xrightarrow{d_{\rho, A}^0} A^4 \xrightarrow{d_{\rho, A}^1} A^2 \longrightarrow 0. \quad (30)$$

Write S, T, X_0, X_1 for the actions of the four lower values on A . With the lift convention (a, g) in a semidirect product, simultaneous infinitesimal conjugation gives

$$d_{\rho, A}^0(v) = ((S - 1)v, (T - 1)v, (X_0 - 1)v, (X_1 - 1)v). \quad (31)$$

The map $d_{\rho, A}^1$ is the ordered pair of the exact tame- and wild-relator defects. Concretely, if the lower map has values in C and a lift to $C \ltimes A$ is written

$$(\rho(\sigma), a), \quad (\rho(\tau), b), \quad (\rho(x_0), c), \quad (\rho(x_1), d),$$

then $d_{\rho, A}^1(a, b, c, d)$ is the coefficient of A in the evaluated tame and wild relators. This construction is functorial in the C -module A because it is obtained by evaluating the same two finite words in semidirect products. Its degree-one kernel is denoted $Z_{A, \rho}^1(A)$ and its cohomology by $H_{A, \rho}^i(A)$.

For $\Gamma = G_{\mathbf{Q}_2}$, put

$$H_{F, \rho}^i(A) = H_{\text{cont}}^i(G_{\mathbf{Q}_2}, A_{\rho}), \quad Z_{F, \rho}^1(A) = Z_{\text{cont}}^1(G_{\mathbf{Q}_2}, A_{\rho}),$$

where the action is induced by ρ . These are the ordinary continuous Galois cohomology groups. No two-relator complex is assigned to $G_{\mathbf{Q}_2}$.

In either theory a finite extension class gives an obstruction in degree two; when it vanishes, the compatible lifts form an affine torsor under the corresponding Z^1 . The comparison below concerns the dimensions, pairings, connecting maps, and quadratic refinements needed by the finite lifting problem, not an identification of the two cochain complexes.

On the candidate side the pro-2 normal-closure condition contributes no additional linear equation in these lifting problems. The elementary kernel A is placed inside the marked normal 2-subgroup of the target; multiplying any of the four chosen lifts by an element of A therefore preserves the marked condition, and the only defects are the two displayed relators.

5.1. The evaluated Fox Jacobian. Let S, T denote the actions of $\rho(\sigma), \rho(\tau)$ on A . If $e = |T|$ is odd, put

$$P = 1 + T + \cdots + T^{e-1}.$$

The following calculation is valid for any finite coefficient module after replacing ω_2 by an integer representative modulo the exponent of the finite semidirect products used in the calculation.

Lemma 5.1 (Finite-exponent independence). *Suppose ν and ν' represent ω_2 modulo the exponent of every finite semidirect and Heisenberg group occurring in a fixed first- or second-order calculation. Replacing every ω_2 -power in the relators by ν or by ν' gives the same first derivative and the same mixed second derivative.*

Proof. The two ordinary powers define the same element in each finite group in which the relator is evaluated. First and mixed derivatives are, by definition, the noncentral and central coordinates of those finite evaluations. They are therefore identical. The representative may be chosen separately for each finite coefficient calculation; no global coherent integer is required. $\square$

Lemma 5.2 (Exact class-two identity). *Let*

$$1 \longrightarrow Z \longrightarrow \tilde{V} \longrightarrow V \longrightarrow 1$$

be a central extension in which V and Z are elementary abelian 2-groups, and let ϕ be an automorphism of $\tilde{V}$ fixing Z . For $X, D \in \tilde{V}$, put

$$\mathfrak{h}_{\phi}(X, D) = \phi(X)X\phi(D)D^2[\phi(D), D].$$

Then

$$\mathfrak{h}_{\phi}(X, D) = \phi(X)XD^{-1}\phi(D). \quad (32)$$

Consequently:

- (i) if D and X have the same image in V , then $\mathfrak{h}_\phi(X, D) = X^2$;
- (ii) if ϕ is the identity, then $\mathfrak{h}_\phi(X, D) = X^2$ for every D .

Proof. The group $\tilde{V}$ has class at most two, every commutator has order at most two, and every element has fourth power one. Put $A = \phi(D)$, $B = D$, and $k = [A, B]$. With our commutator convention $AB = BAK$; since k is central of order at most two and B^2 is central,

$$ABB^2k = BAB^2 = B^3A = B^{-1}A.$$

This proves (32).

If $D = Xz$ with $z \in Z$, then $\phi(z) = z$ and $D^{-1}\phi(D) = X^{-1}\phi(X)$. Hence $\mathfrak{h}_\phi(X, D) = \phi(X)^2 = X^2$, because $X^2 \in Z$ and ϕ fixes Z . If $\phi = 1$, (32) gives the second assertion directly. $\square$

Lemma 5.3 (The elementary and class-two shadow of h_0). *Let V be either the trivial module or a nontrivial simple tame $\mathbf{F}_2[C]$ -module. For an arbitrary lower map, the marked wild subgroup acts trivially on every simple factor, so the evaluated simple-factor derivatives are the same as those of the induced tame lower map. At that tame map:*

- (i) the first derivative of h_0 is the first derivative of x_0^2 , namely zero;
- (ii) on the normalized degree-one representatives supported on x_0 , the mixed Heisenberg and extraspecial central coordinates of h_0 are respectively those of x_0^2 .

Thus h_0 may replace x_0^2 in every linear, cup-product, and determinant calculation used below.

Proof. Let $U = S_2^\omega$. If the first-order variations of τ and x_0 are b and c , then

$$Dd_0 = Pb + (P+1)c, \quad Dx_0 = c.$$

The variation of g_0 affects only terms of order at least two, because the base values of x_0 and d_0 at the tame map are both the identity. If $P = 0$, then $Dd_0 = c = Dx_0$, and the four linear terms in $\mathfrak{h}_{g_0}(x_0, d_0)$ cancel. If $P = 1$, then $T = 1$ and the 2-primary part U acts trivially on a simple characteristic-2 module; again the four linear terms cancel. This proves (i).

For the normalized cohomology representatives used in the mixed and quadratic calculations, only x_0 varies. If $P = 0$, the two elements d_0 and x_0 have the same image c in V , so lemma 5.2(i) applies. If $P = 1$, then $Dd_0 = 0$ and $T = 1$. The 2-primary operator U acts trivially on the simple quotient V . Its lift to a Heisenberg or extraspecial coefficient group is therefore a central automorphism. Such an automorphism has order at most 2, because it is determined by a homomorphism from the elementary quotient to the central involution. Hence $g_0 = \sigma_2^2$ acts trivially on the full coefficient group, and lemma 5.2(ii) applies. The same reasoning covers the trivial module. Taking the mixed Heisenberg central kernel gives the square diagonal $\lambda(c)$; taking the extraspecial central kernel gives $q(c)$. This proves (ii). $\square$

Lemma 5.4 (Full finite-word ledger). *At a tame lower map, write first-order variations of (σ, τ, x_0, x_1) as (a, b, c, d) . Let S, T be the lower actions of σ, τ , put $U = S_2^\omega$, and let $P = 1 + T + \dots + T^{e-1}$ for the odd order e of T . The raw finite Fox rules give*

$$D(u_i) = P(Dx_i + D\tau), \quad D(d_0) = Pb + (P+1)c, \quad D(z_0) = U^{-1}c.$$

The first-order contributions of the wild factors are

factor	a	b	c	d
h_0	0	0	0	0
u_1^{-1}	0	P	0	P
x_1^σ	0	0	0	S^{-1}
$[d_0, z_0]$	0	0	0	0.

Thus the wild row is $L_w = Pb + (P + S^{-1})d$.

For the mixed Heisenberg or extraspecial calculation on the normalized cohomology representative supported on x_0 , the complete central ledger is

factor	V -coordinate	central coordinate
h_0	0	$\lambda(c)$ in $\mathcal{H}(V)$, $q(c)$ for κ_q^0
u_1^{-1}	0	0
x_1^σ	0	0
$d_0 = (x_0\tau)_2^\omega x_0^{-1}$	$(P+1)c$	*
$z_0 = x_0^{\sigma^2}$	$U^{-1}c$	*
$[d_0, z_0]$	0	$B((P+1)c, U^{-1}c)$.

The symbols $*$ are irrelevant because d_0 and z_0 enter the relator only through their commutator. The safety of replacing h_0 by x_0^2 in these ledgers is exactly [lemma 5.3](#): the variation of $g_0 = \sigma_2^2$ starts in order at least two; when $P = 0$ one has $Dd_0 = Dx_0$, so [lemma 5.2\(i\)](#) applies; when $P = 1$, the lift of U to the class-two coefficient group is a central automorphism of order at most two, so g_0 acts trivially and [lemma 5.2\(ii\)](#) applies.

Lemma 5.5 (Evaluated Jacobian). Write the lift variables on (σ, τ, x_0, x_1) as $(a, b, c, d) \in A^4$. On a simple tame module the actual relation differential is

$$d_{\rho, A}^1 = \begin{pmatrix} S^{-1}(1+T) & S^{-1} + 1 + T & 0 & 0 \\ 0 & P & 0 & P + S^{-1} \end{pmatrix}. \quad (33)$$

Equivalently,

$$L_t = S^{-1}(1+T)a + (S^{-1} + 1 + T)b, \quad (34)$$

$$L_w = Pb + (P + S^{-1})d. \quad (35)$$

Postcomposing the tame defect coordinate by S gives the row-normalized expression

$$(1+T)a + (1+S(1+T))b,$$

which may be used for rank calculations only. The word complex, the mixed derivative, and the endpoint maps below use the unnormalized row (34).

Proof. Use $(u, g)(v, h) = (u + gv, gh)$ in $A \rtimes C$. A direct calculation gives

$$D(\tau^\sigma \tau^{-2})(a, b) = S^{-1}(T-1)a + (S^{-1} - (1+T))b.$$

Over $\mathbf{F}_2$ this is exactly (34).

For an odd-order T -orbit, differentiation of a ω_2 -power gives the norm projector P . By [lemma 5.3](#), the word h_0 has the first-order value of x_0^2 , hence zero. The remaining wild factors give

	a	b	c	d
h_0	0	0	0	0
u_1^{-1}	0	P	0	P
x_1^σ	0	0	0	S^{-1}
$[d_0, z_0]$	0	0	0	0.

Their sum is (35). Without the simplicity assumption, the same finite Fox rules define the word complex; the displayed reduction is the simple-factor calculation used in the devissage. $\square$

5.2. The Fox–Heisenberg cup calculation. For an elementary module A , set

$$\mathcal{H}(A) = A \times A^\vee \times \mathbf{F}_2,$$

with multiplication

$$(a, \lambda, z)(a', \lambda', z') = (a + a', \lambda + \lambda', z + z' + \lambda(a')).$$

The C -action is diagonal. If a finite word has evaluated coordinates

$$(\bar{u}, D_u, D_u^\vee, \beta_u),$$

then multiplication and inversion obey

$$D_{uv} = D_u + \bar{u}D_v, \quad D_{uv}^\vee = D_u^\vee + \bar{u}D_v^\vee, \quad (36)$$

$$\beta_{uv} = \beta_u + \beta_v + (D_u^\vee)(\bar{u}D_v), \quad \beta_{u^{-1}} = \beta_u + (\bar{u}^{-1}D_u^\vee)(\bar{u}^{-1}D_u). \quad (37)$$

These are the finite Fox recursion formulas. The Stokes identities below are proved directly by conjugation in the finite Heisenberg group; no cellular diagonal approximation is used. All identities in this subsection are identities of finite word evaluations before cohomology. In particular, coefficient functoriality and the later passage to nonsplit elementary modules are proved at the level of complexes and mapping cones; no semisimple decomposition of the C -module A is used.

Lemma 5.6 (Strict coefficient naturality). *For every C -module map $f : A \rightarrow A'$, the mixed derivatives satisfy*

$$\beta_{\rho,r}^{A'}(f_*a, \lambda') = \beta_{\rho,r}^A(a, f^\vee \lambda').$$

Proof. Evaluate the same finite relator in the mixed Heisenberg group

$$A \times A'^\vee \times \mathbf{F}_2, \quad (a, \lambda, z)(a', \lambda', z') = (a + a', \lambda + \lambda', z + z' + \lambda(f(a'))).$$

Its two natural images in $\mathcal{H}(A')$ and $\mathcal{H}(A)$ have the same central coordinate. This gives the identity before passage to cohomology. $\square$

Lemma 5.7 (Finite-word Stokes formula). *Let $F_n = \langle g_1, \dots, g_n \rangle$ be a free group, let $\rho : F_n \rightarrow C$ be a homomorphism to a finite group, and write $c_i = \rho(g_i)$. Let $r \in F_n$ be an ordinary finite word with $\rho(r) = 1$, and let $\epsilon_i(r) \in \mathbf{F}_2$ be the total exponent of g_i in r , reduced modulo 2.*

For $x = (x_i) \in A^n$ and $y = (y_i) \in (A^\vee)^n$, evaluate r after replacing g_i by $(x_i, y_i, 0; c_i) \in \mathcal{H}(A) \rtimes C$. Write the result as

$$(L_r^A(x), L_r^{A^\vee}(y), \beta_r(x, y); 1).$$

Then, for $a \in A$ and $\lambda \in A^\vee$,

$$\beta_r(d^0 a, y) = \langle a, L_r^{A^\vee}(y) \rangle + \sum_{i=1}^n \epsilon_i(r) y_i(c_i a), \quad (38)$$

$$\beta_r(x, d^0 \lambda) = \langle L_r^A(x), \lambda \rangle + \sum_{i=1}^n \epsilon_i(r) \lambda(x_i), \quad (39)$$

where

$$d^0 a = ((c_i - 1)a)_i, \quad d^0 \lambda = ((c_i - 1)\lambda)_i.$$

Consequently, for words $r_1, \dots, r_m$ and coefficients $t_j \in \mathbf{F}_2$ satisfying

$$\sum_{j=1}^m t_j \epsilon_i(r_j) = 0 \quad (1 \leq i \leq n), \quad (40)$$

the corresponding traced mixed coordinate satisfies the two Stokes identities without correction terms.

Proof. Put

$$p_a = (a, 0, 0; 1), \quad p_\lambda = (0, \lambda, 0; 1), \quad \mathbf{z}(t) = (0, 0, t; 1).$$

A direct multiplication in $\mathcal{H}(A) \rtimes C$ gives

$$\begin{aligned} p_a^{-1}(0, y_i, 0; c_i) p_a &= ((c_i - 1)a, y_i, 0; c_i) \mathbf{z}(y_i(c_i a)), \\ p_\lambda^{-1}(x_i, 0, 0; c_i) p_\lambda &= (x_i, (c_i - 1)\lambda, 0; c_i) \mathbf{z}(\lambda(x_i)). \end{aligned}$$

The elements $\mathbf{z}(t)$ are central of order at most two. Therefore, on evaluating an ordinary word, the extra central factor attached to the i th generator occurs exactly $\epsilon_i(r)$ times.

If only the dual variables y_i are present, the value of r is $(0, L_r^{A^\vee}(y), 0; 1)$. Conjugating this value by p_a contributes the central coordinate $L_r^{A^\vee}(y)(a)$. Adding the generatorwise central corrections gives (38). The same argument with p_λ starts from $(L_r^A(x), 0, 0; 1)$ and gives (39). Condition (40) cancels the correction terms. Every step is functorial in A . $\square$

Proposition 5.8 (Traced finite-word Stokes identity). *Let β_t and β_w be the mixed central coordinates obtained from the tame and wild relators, and set*

$$\mathcal{B}_{\rho,A} = \beta_t + \beta_w.$$

Let L_t and L_w denote the two unnormalized first relation differentials. Then, before passage to cohomology,

$$\mathcal{B}_{\rho,A}(d_{\rho,A}^0 a, y) = \langle a, L_t^{A^\vee}(y) + L_w^{A^\vee}(y) \rangle, \quad (41)$$

$$\mathcal{B}_{\rho,A}(x, d_{\rho,A^\vee}^0 \lambda) = \langle L_t^A(x) + L_w^A(x), \lambda \rangle. \quad (42)$$

The identities are natural in A . They are assertions about the sum of the two relator coordinates; in general they are false for either relator separately.

Proof. Choose an odd ordinary integer representative of ω_2 which is valid in every finite semidirect-Heisenberg group in the calculation, and replace the two profinite relators by the resulting ordinary words r_t and r_w . In the ordered generator list (σ, τ, x_0, x_1) their mod-2 exponent vectors are

$$\epsilon(r_t) = (0, 1, 0, 0), \quad \epsilon(r_w) = (0, 1, 0, 0).$$

Indeed, $r_t = \tau^\sigma \tau^{-2}$ has odd total τ -exponent. For $r_w = h_0 u_1^{-1} x_1^\sigma c_0$, the commutator c_0 has zero exponent vector. The word h_0 also has zero exponent vector: the two x_0 -terms cancel modulo 2, the two occurrences of d_0 cancel, d_0^2 has even exponent vector, and h_c is a commutator. Finally, u_1^{-1} contributes $x_1 + \tau$ and x_1^σ contributes x_1 . Thus the sum of the two exponent vectors is zero.

Apply lemma 5.7 with trace coefficients $(1, 1)$. The correction terms in (38) cancel and give (41); the correction terms in (39) cancel and give (42). Independence of the chosen integer representative is lemma 5.1, and strict coefficient naturality is lemma 5.6. $\square$

Remark 5.9 (The trace is essential). *There is a small explicit regression test. Take $C = \mathrm{GL}_2(\mathbf{F}_2)$ with*

$$S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad T = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \quad S^{-1}TS = T^2,$$

and let $A = \mathbf{F}_2^2$ be the natural module. For $a = e_2$ and the dual variation supported on τ with value $e_2^\vee$, the tame relator alone has mixed central value 0, whereas pairing its dual first derivative with a gives 1. The wild relator supplies the compensating value. Thus a relator-by-relator Stokes formula would assert $0 = 1$, while the traced formula above is exact.

For $x \in A^4$ and $y \in (A^\vee)^4$, let

$$\mathcal{B}_{\rho,A}(x, y) = \beta_{\rho,r_t}(x, y) + \beta_{\rho,r_w}(x, y)$$

be the sum of the two mixed central coordinates. The following proposition constructs the chain map used below, including its degree-zero and degree-two components.

Proposition 5.10 (The Fox–Heisenberg chain map). *Under the canonical identifications*

$$\mathrm{Hom}((A^\vee)^2, \mathbf{F}_2) \cong A^2, \quad \mathrm{Hom}((A^\vee)^4, \mathbf{F}_2) \cong A^4, \quad \mathrm{Hom}(A^\vee, \mathbf{F}_2) \cong A,$$

define

$$D_{\rho,A}^0(a) = (a, a), \quad (43)$$

$$\langle D_{\rho,A}^1(x), y \rangle = \mathcal{B}_{\rho,A}(x, y), \quad (44)$$

$$D_{\rho,A}^2(u_t, u_w) = u_t + u_w. \quad (45)$$

Then $D_{\rho,A} = (D^0, D^1, D^2)$ is a natural chain map

$$D_{\rho,A} : C_{\rho,A}^\bullet \longrightarrow \mathrm{Hom}_{\mathbf{F}_2}(C_{\rho,A^\vee}^\bullet, \mathbf{F}_2)[-2]. \quad (46)$$

More explicitly,

$$D_{\rho,A}^1 d_{\rho,A}^0 = (d_{\rho,A^\vee}^1)^\vee D_{\rho,A}^0, \quad (47)$$

$$D_{\rho,A}^2 d_{\rho,A}^1 = (d_{\rho,A^\vee}^0)^\vee D_{\rho,A}^1. \quad (48)$$

Proof. The endpoint maps are induced by the scalar trace $\mathrm{tr}(y_t, y_w) = y_t + y_w$. Equations (41) and (42), proved by the direct finite-word calculation of lemma 5.7, are exactly (47) and (48), respectively. Strict naturality in A is lemma 5.6 and proposition 5.8. Hence (46) is a natural chain map in all three degrees. $\square$

Lemma 5.11 (Exact cone dévissage). *For an elementary $\mathbf{F}_2[C]$ -module A , put*

$$K(A) = \mathrm{Cone}\left(D_{\rho,A} : C_{\rho,A}^\bullet \longrightarrow \mathrm{Hom}_{\mathbf{F}_2}(C_{\rho,A^\vee}^\bullet, \mathbf{F}_2)[-2]\right). \quad (49)$$

For every short exact sequence of finite $\mathbf{F}_2[C]$ -modules

$$0 \longrightarrow A' \longrightarrow A \longrightarrow A'' \longrightarrow 0,$$

there is a degreewise short exact sequence of complexes

$$0 \longrightarrow K(A') \longrightarrow K(A) \longrightarrow K(A'') \longrightarrow 0. \quad (50)$$

Consequently, if two of $K(A')$, $K(A)$, and $K(A'')$ are acyclic, then so is the third. In particular, acyclicity for all simple modules implies acyclicity for every elementary module, including nonsplit and nonsemisimple modules.

Proof. The finite-word construction is additive and exact in the coefficient module, so the source complexes form a degreewise short exact sequence

$$0 \longrightarrow C_{\rho,A'}^\bullet \longrightarrow C_{\rho,A}^\bullet \longrightarrow C_{\rho,A''}^\bullet \longrightarrow 0. \quad (51)$$

Dualizing the coefficient sequence gives

$$0 \longrightarrow A''^\vee \longrightarrow A^\vee \longrightarrow A'^\vee \longrightarrow 0.$$

Applying the same finite-word complex and then $\mathrm{Hom}_{\mathbf{F}_2}(-, \mathbf{F}_2)$ reverses this sequence once more, and yields

$$\begin{aligned} 0 \longrightarrow \mathrm{Hom}_{\mathbf{F}_2}(C_{\rho,A'^\vee}^\bullet, \mathbf{F}_2)[-2] &\longrightarrow \mathrm{Hom}_{\mathbf{F}_2}(C_{\rho,A^\vee}^\bullet, \mathbf{F}_2)[-2] \\ &\longrightarrow \mathrm{Hom}_{\mathbf{F}_2}(C_{\rho,A''^\vee}^\bullet, \mathbf{F}_2)[-2] \longrightarrow 0. \end{aligned} \quad (52)$$

For compactness write

$$C^\dagger(A) = \mathrm{Hom}_{\mathbf{F}_2}(C_{\rho,A^\vee}^\bullet, \mathbf{F}_2)[-2].$$

Strict coefficient naturality of the Heisenberg central coordinate gives the commutative diagram

$$\begin{array}{ccccc} C_{\rho,A'}^\bullet & \longrightarrow & C_{\rho,A}^\bullet & \longrightarrow & C_{\rho,A''}^\bullet \\ \downarrow D_{\rho,A'} & & \downarrow D_{\rho,A} & & \downarrow D_{\rho,A''} \\ C^\dagger(A') & \longrightarrow & C^\dagger(A) & \longrightarrow & C^\dagger(A''). \end{array}$$

Here the rows are the nonzero portions of (51) and (52), and commutativity is lemma 5.6 and proposition 5.10. Mapping cones are formed degree by degree, so this diagram gives (50). The associated long exact cohomology sequence proves the two-out-of-three assertion. Applying it successively to a composition series of A proves the final statement. $\square$

Lemma 5.12 (Simple characteristic-two modules are tame). *Let V be a simple $\mathbf{F}_2[C]$ -module occurring in the induction. Then the marked normal 2-subgroup of C acts trivially on V . Hence V factors through the finite tame quotient of C .*

Proof. A finite 2-group acting on a nonzero vector space over $\mathbf{F}_2$ has a nonzero fixed vector. Since the 2-subgroup is normal in C , the fixed subspace is a nonzero C -submodule. Simplicity forces this fixed subspace to be all of V . $\square$

Lemma 5.13 (Normal forms for nontrivial simple modules). *Let V be a nontrivial simple tame $\mathbf{F}_2[C]$ -module. Every class in $H_{A,\rho}^1(V)$ has a unique representative supported on x_0 :*

$$(a, b, c, d) \sim (0, 0, c, 0). \quad (53)$$

More precisely, the two possible inertia cases are as follows.

(i) If $T = 1$, then $P = 1$, $1 + S^{-1}$ is invertible, and

$$Z_{A,\rho}^1(V) = \{(a, 0, c, 0) : a, c \in V\}, \quad B_{A,\rho}^1(V) = \{((S - 1)v, 0, 0, 0) : v \in V\}.$$

(ii) If $V^T = 0$, then $P = 0$ and $T - 1 = 1 + T$ is invertible. The wild row first forces $d = 0$. Subtracting the coboundary of $v = (T - 1)^{-1}b$ kills b , and the tame row then forces $a = 0$.

Under the representatives (53), the degree-one pairing with the corresponding representative $\lambda \in V^\vee$ is

$$(c, \lambda) \mapsto \begin{cases} \lambda(c), & T = 1, \\ \lambda((1 + U + U^{-1})c), & V^T = 0. \end{cases} \quad (54)$$

Both operators in (54) are invertible.

Proof. Because tame inertia is normal, V^T is a C -submodule; simplicity gives either $T = 1$ and $P = 1$, or $V^T = 0$ and $P = 0$.

If $T = 1$, the two rows of (33) reduce to $b = 0$ and $(1 + S^{-1})d = 0$. A nonzero kernel of $1 + S^{-1}$ would be the S -fixed subspace and hence a nonzero submodule, so $1 + S^{-1}$ is invertible and $d = 0$. The displayed descriptions of Z^1 and B^1 follow. The same argument shows that $S - 1$ is invertible, so each class has a unique representative $(0, 0, c, 0)$.

If $V^T = 0$, then $1 + T$ is invertible. The wild row is $S^{-1}d = 0$, hence $d = 0$. For a cocycle $(a, b, c, 0)$, subtract the coboundary of $v = (T - 1)^{-1}b$. The new representative has $b = 0$, and its tame row is $S^{-1}(1 + T)a = 0$, so $a = 0$. Uniqueness follows because a coboundary with zero T -coordinate has $(T - 1)v = 0$, hence $v = 0$.

The pairing formula is lemma 5.14. In the ramified case U has 2-power order, so its minimal polynomial divides a power of $X + 1$. Since $X^2 + X + 1$ is coprime to $X + 1$,

$$1 + U + U^{-1} = U^{-1}(U^2 + U + 1)$$

is invertible. The unramified operator is the identity. $\square$

Lemma 5.14 (Mixed Hessian calculation). *Let V be a nontrivial simple tame module, put $U = S_2^\omega$, and represent a class in $H_{A,\rho}^1(V)$ by the cocycle supported on x_0 with value c . Represent the dual class similarly by $\lambda \in V^\vee$. The mixed central contributions are*

word contribution	$T = 1$	$V^T = 0$
r_t	0	0
h_0	$\lambda(c)$	$\lambda(c)$
$[d_0, z_0]$	0	$\lambda(Uc) + \lambda(U^{-1}c)$
all other factors of r_w	0	0.

Consequently the degree-one pairing is induced by

$$\begin{cases} 1, & T = 1, \\ 1 + U + U^{-1}, & V^T = 0. \end{cases} \quad (55)$$

Proof. The representative exists by the row reduction of (33). By lemma 5.3, the word h_0 contributes exactly the usual Heisenberg square diagonal $\lambda(c)$. In the ramified case the first derivatives on the x_0 -coordinate are

$$Dd_0(c) = c, \quad Dz_0(c) = U^{-1}c,$$

whereas on the dual coordinate they are

$$Dd_0(\lambda) = \lambda, \quad Dz_0(\lambda) = U^\vee \lambda.$$

Applying (36)–(37) to the commutator gives

$$\lambda(U^{-1}c) + (U^\vee \lambda)(c) = \lambda((U^{-1} + U)c).$$

The remaining factors have either zero first derivative on the normalized representatives or no mixed central term. Adding the h_0 contribution proves the table. $\square$

Proposition 5.15 (Candidate deformation duality). *For every elementary $\mathbf{F}_2[C]$ -module A , the chain map (46) is a quasi-isomorphism. Hence*

$$H_{A,\rho}^2(A)^\vee \cong (A^\vee)^C, \quad \dim Z_{A,\rho}^1(A) = 2 \dim A + \dim (A^\vee)^C. \quad (56)$$

The induced pairings

$$H_{A,\rho}^i(A) \times H_{A,\rho}^{2-i}(A^\vee) \rightarrow \mathbf{F}_2$$

are perfect and natural in A .

Proof. For $A = \mathbf{F}_2$, all lower actions are trivial, so $d^0 = 0$ and the actual Jacobian is

$$d^1(a, b, c, d) = (b, b).$$

Thus $\dim H^0 = 1$, $\dim H^1 = 3$, and $\dim H^2 = 1$. On the basis of H^1 represented by the coordinates (σ, x_0, x_1) , the traced Fox–Heisenberg form is

$$((a, c, d), (a', c', d')) \mapsto ad' + da' + cc',$$

with matrix

$$\begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}.$$

It is nonsingular, and the endpoint trace gives a nonzero H^0 – H^2 pairing. This is exactly the scalar cup–Bockstein table (25).

For a nontrivial simple module, lemma 5.12 makes the action tame, and lemma 5.13 gives a unique degree-one representative supported on x_0 . The Jacobian is surjective, so $H^0 = H^2 = 0$ and $\dim H^1 = \dim A$. On the unique normal forms the degree-one pairing is exactly (54), and is nonsingular by the final assertion of lemma 5.13. Hence the cone $K(A)$ is acyclic for every simple module, including the trivial simple module treated above.

For a general elementary module, choose a composition series. The degreewise short exact cone sequence (50) and its long exact cohomology sequence give acyclicity at each extension step. Thus $D_{\rho,A}$ is a quasi-isomorphism for every elementary, possibly nonsplit and nonsemisimple, module A . The numerical formula follows from duality and the Euler characteristic of (30). $\square$

Proposition 5.16 (Local lifting duality). *For every elementary $\mathbf{F}_2[C]$ -module A ,*

$$H_{F,\rho}^2(A)^\vee \cong (A^\vee)^C, \quad \dim Z_{F,\rho}^1(A) = 2 \dim A + \dim (A^\vee)^C. \quad (57)$$

The local cup product and invariant map give perfect pairings

$$H_{F,\rho}^i(A) \times H_{F,\rho}^{2-i}(A^\vee) \rightarrow \mathbf{F}_2.$$

Proof. This is local Tate duality, with the mod-2 cyclotomic twist trivial, together with the local Euler–Poincaré formula for $\mathbf{Q}_2$; see [1, Chapter VII, Sections 2–3]. Since the action factors through C , the invariant space is $(A^\vee)^C$. Finally

$$\dim Z^1 = \dim H^1 + \dim B^1 = \dim H^1 + \dim A - \dim A^C,$$

and the Euler characteristic gives the stated expression. $\square$

The next corollary is the only comparison between the two theories used later. It compares functorial invariants; it does not identify the two H^1 -spaces.

Corollary 5.17 (Common lift numerics and adjoint boundaries). *For either source, use the corresponding notation $H_{\Gamma, \rho}^i$ and $Z_{\Gamma, \rho}^1$. Then the obstruction-space dimensions, unobstructed lift multiplicities, and perfect pairings agree numerically.*

For every short exact sequence

$$0 \rightarrow T \rightarrow M \rightarrow V \rightarrow 0$$

and its dual, the connecting maps are adjoint: if

$$\partial_{\Gamma, \rho} : H_{\Gamma, \rho}^1(V) \rightarrow H_{\Gamma, \rho}^2(T)$$

and $\chi \in (T^\vee)^C$ has dual boundary

$$\gamma_\chi = \partial_C^\vee(\chi) \in H^1(C, V^\vee),$$

then

$$\langle \partial_{\Gamma, \rho} c, \chi \rangle = \langle c, \rho^* \gamma_\chi \rangle_\Gamma. \quad (58)$$

Proof. For the candidate, (58) is the naturality of the chain map (46) applied to the coefficient sequence. For the local source it is the standard functoriality of Tate duality. The numerical assertions are [propositions 5.15](#) and [5.16](#). $\square$

6. QUADRATIC DETERMINANT OBSTRUCTIONS

For the candidate source we write

$$\iota_A : H_{A, \rho}^2(\mathbf{F}_2) \longrightarrow \mathbf{F}_2$$

for the functional induced by the degree-two trace

$$D^2(u_t, u_w) = u_t + u_w$$

of [proposition 5.10](#). For the local source we write

$$\iota_F = \text{inv}_{\mathbf{Q}_2} : H^2(G_{\mathbf{Q}_2}, \mathbf{F}_2) \longrightarrow \mathbf{F}_2.$$

All quadratic obstruction functions below use the source-specific functional ι_Γ .

The determinant comparison is made with *equivariant* central extension classes. A C -invariant quadratic form on V does not in general admit a C -invariant factor set on the chosen section $V \rightarrow E$; the central correction terms in the lifted C -action are part of the data. We therefore separate a fixed base equivariant determinant class from the target-side affine data which will later be handled by phase covers.

Lemma 6.1 (Equivariant extraspecial connecting cocycle). *Let V be an elementary $\mathbf{F}_2[C]$ -module, let $q : V \rightarrow \mathbf{F}_2$ be a C -invariant quadratic form with polar form B , and let*

$$f : V \times V \rightarrow \mathbf{F}_2, \quad f(v, v) = q(v), \quad f(v, w) + f(w, v) = B(v, w),$$

be a normalized factor set for the central extension

$$E_f = V \times \mathbf{F}_2, \quad (v, z)(w, t) = (v + w, z + t + f(v, w)).$$

When an equivariant lift is chosen, its action in the normalized section is written by functions $m_c : V \rightarrow \mathbf{F}_2$ satisfying

$$m_c(v + w) + m_c(v) + m_c(w) = f(cv, cw) + f(v, w) \quad (59)$$

and

$$m_{cd}(v) = m_c(dv) + m_d(v), \quad m_1 = 0. \quad (60)$$

Equivalently,

$$c \cdot (v, z) = (cv, z + m_c(v))$$

defines an action by automorphisms of E_f . The associated normalized base central cocycle on $V \rtimes C$ is

$$\kappa_q^0((v, c), (w, d)) = f(v, cw) + m_c(w). \quad (61)$$

Consequently, for a source Γ , a lower map $\rho : \Gamma \rightarrow C$, and a normalized V -valued 1-cocycle b , the base connecting cocycle is

$$(b, \rho)^* \kappa_q^0(g, h) = f(b(g), \rho(g)b(h)) + m_{\rho(g)}(b(h)). \quad (62)$$

If κ is another normalized central cocycle on $V \rtimes C$ whose restriction to V is the same quadratic form q , then, after changing normalized sections, its difference from the fixed base class has the form

$$\kappa = \kappa_q^0 + \Gamma_\gamma + \inf_C^{V \rtimes C} \delta, \quad (63)$$

where

$$\Gamma_\gamma((v, c), (w, d)) = \gamma(c)(cw), \quad \gamma \in Z^1(C, V^\vee), \quad \delta \in Z^2(C, \mathbf{F}_2). \quad (64)$$

The corresponding quadratic obstruction is

$$Q_{\kappa, \Gamma, \rho}(b) = Q_{\Gamma, \rho}^0(b) + \langle b, \rho^* \gamma \rangle_\Gamma + \iota_\Gamma(\rho^* \delta), \quad (65)$$

where

$$Q_{\Gamma, \rho}^0(b) = \iota_\Gamma((b, \rho)^* \kappa_q^0).$$

Its polarization is the source pairing transported by B .

Proof. Equation (59) is exactly the condition that $(v, z) \mapsto (cv, z + m_c(v))$ preserve the multiplication in E_f ; equation (60) says that these automorphisms form a genuine C -action. Multiplication in the semidirect product attached to this lifted action is

$$(v, c, z)(w, d, t) = (v + cw, cd, z + t + f(v, cw) + m_c(w)), \quad (66)$$

which gives (61). Pulling back this cocycle by the graph of a 1-cocycle b gives (62). This is the term which would be missing if one incorrectly assumed that the section $v \mapsto (v, 0)$ were C -equivariant.

For the difference statement, the quotient $\kappa - \kappa_q^0$ restricts trivially to $V \times V$. In the normalized Lyndon–Hochschild–Serre filtration for the split extension $1 \rightarrow V \rightarrow V \rtimes C \rightarrow C \rightarrow 1$, such a class has only a filtration-one part in $Z^1(C, V^\vee)$ and an inflated scalar part in $Z^2(C, \mathbf{F}_2)$. The normalized representative is exactly (64); the cocycle condition for it is the condition that γ be a 1-cocycle. Pullback along (b, ρ) gives (65). Polarization of the base term is obtained by adding the formulas for $b + b'$, b , and b' ; the m_c -contribution cancels because each m_c has polar form $f(c-, c-) + f(-, -)$, and the remaining term is $B(b \smile b')$. The difference terms in (65) are linear or inflated scalar terms and therefore do not change the polar form. $\square$

Lemma 6.2 (Explicit involution-orbit cocycle). *Let H be a finite group, let $g \in H$ have order 2, and let $M = \mathbf{F}_2[H]$ with left-regular action*

$$(cx)_h = x_{c^{-1}h}.$$

Choose a set $\mathcal{R} \subset H$ of representatives for the right cosets $u\langle g \rangle$. For $c \in H$ and $u \in \mathcal{R}$, write uniquely

$$c^{-1}u = \pi_c(u)g^{\epsilon_c(u)}, \quad \pi_c(u) \in \mathcal{R}, \quad \epsilon_c(u) \in \mathbf{F}_2. \quad (67)$$

Define

$$q_g(x) = \sum_{u \in \mathcal{R}} x_u x_{ug}, \quad (68)$$

$$f_g(x, y) = \sum_{u \in \mathcal{R}} x_u y_{ug}, \quad (69)$$

$$m_c^g(x) = \sum_{u \in \mathcal{R}} \epsilon_c(u) x_{\pi_c(u)} x_{\pi_c(u)g}. \quad (70)$$

Then

$$m_c^g(x + y) + m_c^g(x) + m_c^g(y) = f_g(cx, cy) + f_g(x, y), \quad (71)$$

$$m_{cd}^g(x) = m_c^g(dx) + m_d^g(x), \quad m_1^g = 0. \quad (72)$$

Consequently

$$\kappa_g^{\mathcal{R}}((x, c), (y, d)) = f_g(x, cy) + m_c^g(y) \quad (73)$$

is a normalized 2-cocycle on $M \rtimes H$. It vanishes on the zero section $H \hookrightarrow M \rtimes H$, and its restriction to M has square map q_g .

Proof. From (67) one obtains

$$\pi_{cd} = \pi_d \circ \pi_c, \quad \epsilon_{cd}(u) = \epsilon_c(u) + \epsilon_d(\pi_c(u)). \quad (74)$$

Moreover,

$$f_g(cx, cy) = \sum_{u \in \mathcal{R}} x_{\pi_c(u)g^{\epsilon_c(u)}} y_{\pi_c(u)g^{\epsilon_c(u)+1}}.$$

For $\epsilon_c(u) = 0$ this is the summand of $f_g(x, y)$ indexed by $\pi_c(u)$; for $\epsilon_c(u) = 1$ its orientation is reversed. Therefore

$$f_g(cx, cy) + f_g(x, y) = \sum_{u \in \mathcal{R}} \epsilon_c(u) (x_{\pi_c(u)} y_{\pi_c(u)g} + x_{\pi_c(u)g} y_{\pi_c(u)}),$$

which is exactly the polarization of (70). This proves (71).

The product $x_v x_{vg}$ is unchanged when v is replaced by vg . Hence

$$m_c^g(dx) = \sum_{u \in \mathcal{R}} \epsilon_c(u) x_{\pi_d(\pi_c(u))} x_{\pi_d(\pi_c(u))g}.$$

Adding $m_d^g(x)$ and using (74) gives (72). The cocycle assertion now follows directly from lemma 6.1. Finally, $f_g(x, x) = q_g(x)$ and $m_1^g = 0$, proving the restriction and normalization claims. $\square$

Lemma 6.3 (Base equivariant determinant class). *Let V be a nontrivial simple self-dual tame $\mathbf{F}_2[C]$ -module, let $H = H_V$ be its faithful tame image, and let $q : V \rightarrow \mathbf{F}_2$ be a nonzero H -invariant nonsingular quadratic form. There is a normalized base class*

$$\kappa_q^0 \in Z^2(V \rtimes H, \mathbf{F}_2)$$

whose restriction to V has square map q .

Choose an H -split pair

$$V \xrightarrow{i} W = \mathbf{F}_2[H]^N \xrightarrow{p} V, \quad pi = 1,$$

and put $q_W = q \circ p$. In the left-regular coordinates $X_{j,h}$ of W , the unique algebraic normal form of the H -invariant quadratic map q_W is a sum of orbit polynomials of the following

three types:

$$\begin{aligned} S_j &= \sum_{h \in H} X_{j,h}^2, \\ C_{j,k,g} &= \sum_{h \in H} X_{j,h} X_{k,hg} \quad \text{when the corresponding unordered pair has trivial stabilizer,} \\ E_{j,g} &= \sum_{u \in \mathcal{R}_g} X_{j,u} X_{j,ug}, \quad g^2 = 1, \quad g \neq 1, \end{aligned}$$

where $\mathcal{R}_g$ is any set of representatives for the right cosets of $\langle g \rangle$ in H . For a square orbit use the H -invariant factor set

$$f_{S_j}(x, y) = \sum_{h \in H} x_{j,h} y_{j,h}, \quad m_c = 0. \quad (75)$$

For a free orbit use

$$f_{C_{j,k,g}}(x, y) = \sum_{h \in H} x_{j,h} y_{k,hg}, \quad m_c = 0. \quad (76)$$

For an involution-stabilized orbit use the cocycle (73) of lemma 6.2 on the j th regular summand, with $\mathcal{R} = \mathcal{R}_g$. Let $\kappa_{q_W}^0$ be the sum of the cocycles corresponding to the orbit polynomials occurring in q_W , and put

$$\kappa_q^0 = (i \times 1)^* \kappa_{q_W}^0. \quad (77)$$

Then κ_q^0 vanishes on the zero section and $\text{res}_V(\kappa_q^0)$ has square map q . If a lower target has quotient C acting through H , we inflate this class along $C \rightarrow H$ and keep the same notation.

Proof. If V is ramified, V is projective over $\mathbf{F}_2[H]$ by lemma 6.11; if V is unramified, the 2-primary subgroup of the cyclic unramified image acts trivially on every simple characteristic-2 module, so the faithful image H has odd order and Maschke's theorem applies. Thus the split pair exists.

The algebraic normal form of a quadratic map over $\mathbf{F}_2$ is determined by its values on the coordinate vectors and by its polar form. H -invariance makes its coefficients constant on the orbits of unordered degree-two coordinate monomials. Since the action on each regular coordinate set is free, the stabilizer of an unordered pair is either trivial or has order two and exchanges the two entries. The latter possibility is exactly $j = k$ with a nontrivial involution g , and gives $E_{j,g}$.

The factor sets (75) and (76) are H -invariant, vanish on the zero section, and have square maps S_j and $C_{j,k,g}$, respectively. By lemma 6.2, the explicit cocycle for $E_{j,g}$ has square map $E_{j,g}$ and also vanishes on the zero section. Their sum is therefore an actual normalized equivariant central cocycle on $W \rtimes H$ with fibre square map q_W . Pullback along $i \times 1$ gives fibre square map $q_W(iv) = q(v)$ and preserves zero-section normalization. $\square$

Lemma 6.4 (Uniqueness and isometry invariance of the normalized base class). *Let V , $H = H_V$, and q be as in lemma 6.3. Then*

$$H^1(H, V^\vee) = 0. \quad (78)$$

Any two classes in $H^2(V \rtimes H, \mathbf{F}_2)$ which restrict to the class with square map q on V and restrict trivially along the zero section $H \hookrightarrow V \rtimes H$ are equal. Consequently, changing the normalized factor set, the correction functions m_c , the half-orbit transversals, or the H -split regular embedding and retraction changes κ_q^0 only by a normalized coboundary. In particular the base quadratic obstruction $Q_{\Gamma, \rho}^0$ is well-defined.

Moreover, every H -equivariant isometry u of (V, q) fixes the normalized base class:

$$(u \times 1)^*[\kappa_q^0] = [\kappa_q^0]. \quad (79)$$

Proof. If V is unramified, its faithful image H has odd order: the 2-primary subgroup of the cyclic unramified image acts trivially on every simple characteristic-2 module. Hence positive-degree $\mathbf{F}_2[H]$ -cohomology vanishes by averaging.

Assume that V is ramified, and let $I \triangleleft H$ be the odd tame-inertia subgroup. The fixed space $(V^\vee)^I$ is an H -submodule. It is zero, because V is simple and inertia acts nontrivially. Since $|I|$ is odd, $H^j(I, V^\vee) = 0$ for $j > 0$. The Hochschild–Serre spectral sequence for $1 \rightarrow I \rightarrow H \rightarrow H/I \rightarrow 1$ therefore gives [5, Chapter VII]

$$H^n(H, V^\vee) = 0 \quad (n \geq 0).$$

In particular (78) holds in both cases.

Put $G = V \rtimes H$. For an elementary abelian 2-group V , the square-map correspondence

$$H^2(V, \mathbf{F}_2) \xrightarrow{\sim} \{\text{quadratic maps } V \rightarrow \mathbf{F}_2\}$$

is an isomorphism: the polarization recovers the commutator form, and the square and commutator data determine the central extension class. Thus two classes with square map q have the same fibre restriction.

The low-degree Lyndon–Hochschild–Serre sequence for $1 \rightarrow V \rightarrow G \rightarrow H \rightarrow 1$ contains the exact segment [5, Chapter VII]

$$H^2(H, \mathbf{F}_2) \xrightarrow{\inf} \ker\left(H^2(G, \mathbf{F}_2) \longrightarrow H^2(V, \mathbf{F}_2)\right) \xrightarrow{\text{edge}} H^1(H, V^\vee). \quad (80)$$

Let ξ be the difference of two classes with the stated normalizations. Its fibre restriction is zero. By (78) and exactness of (80), $\xi = \inf(\delta)$ for some $\delta \in H^2(H, \mathbf{F}_2)$. Pullback along the zero section is a left inverse to inflation, while both classes vanish there. Hence $\delta = 0$ and $\xi = 0$.

At the filtration-one level one can see the same cancellation directly. If $\gamma = dh$ for $h \in V^\vee$, then the normalized 1-cochain

$$\ell_h(v, c) = h(v) \quad (81)$$

on $V \rtimes H$ satisfies $d\ell_h = \Gamma_{dh}$, with Γ as in (64).

If normalized cocycle representatives are used, their difference is da for a normalized 1-cochain a . Since the difference vanishes on $H \times H$, $a|_H$ is a 1-cocycle. Subtracting its inflation makes a vanish on the zero section without changing da . Thus all choices in lemma 6.3 give isomorphic normalized central extensions, and transport of lifts through that isomorphism leaves both the candidate relator obstruction and the local pullback class unchanged. This proves the choice-independence of $Q_{\Gamma, \rho}^0$.

Finally, $(u \times 1)^* \kappa_q^0$ has the same fibre square map and the same zero-section restriction as κ_q^0 , so uniqueness proves (79).

The uniqueness assertion is deliberately made over the faithful image H_V . After inflation to a larger quotient C acting through H_V , the group $H^1(C, V^\vee)$ need not vanish; its contribution is precisely the filtration-one term Γ_γ in (63), and it remains part of the target-side affine data. $\square$

6.1. The exact candidate base quadratic form. Let V be a nontrivial self-dual simple tame module, let $q : V \rightarrow \mathbf{F}_2$ be invariant and nonsingular, let B be its polar form, and let κ_q^0 be the base equivariant determinant class of lemma 6.3. The candidate determinant calculation below is made in the lifted semidirect product (66); thus all m_c -terms are included.

Proposition 6.5 (Complete base second-order word expansion). *With $S, T, P, U = S_2^\omega$ as above, evaluation of the actual relator (6) for the base class κ_q^0 gives*

$$Q_A^0(c) = q(c) + B((P+1)c, U^{-1}c). \quad (82)$$

Equivalently,

$$Q_A^0(c) = \begin{cases} q(c), & T = 1, \\ q(c) + B(c, U^{-1}c), & V^T = 0. \end{cases} \quad (83)$$

Its ramified polar form is

$$B_A(c, c') = B(c, (1 + U + U^{-1})c'). \quad (84)$$

For a general determinant class $\kappa = \kappa_q^0 + \Gamma_{\gamma_\kappa} + \inf \delta_\kappa$, one has

$$Q_{A, \kappa, \rho}(c) = Q_A^0(c) + \langle c, \rho^* \gamma_\kappa \rangle_A + \iota_A(\rho^* \delta_\kappa). \quad (85)$$

Proof. Normalize a cohomology representative so that only x_0 varies, by c . In terms of (66), the complete base contribution ledger is

factor	V-coordinate	central coordinate
h_0	0	$q(c)$
u_1^{-1}	0	0
x_1^σ	0	0
$d_0 = (x_0 \tau)_2^\omega x_0^{-1}$	$(P+1)c$	*
$z_0 = x_0^{\sigma_2^\omega}$	$U^{-1}c$	*
$[d_0, z_0]$	0	$B((P+1)c, U^{-1}c)$

The starred entries contain ordinary f -terms and the m -terms of the lifted C -action. They occur only as central coordinates of d_0 and z_0 , and central coordinates do not affect the commutator. Since the lower values of d_0 and z_0 at the tame map are both the identity, the commutator contribution is exactly the polar form of their V -coordinates, $B((P+1)c, U^{-1}c)$.

The first line is the extraspecial case of lemma 5.3, but we recall why it remains valid with the m -terms present. The class-two identity lemma 5.2 uses only that the lift of the relevant tame operator is an automorphism of the central extension and fixes the centre; it does not assume that the chosen section is equivariant. If $P = 0$, then d_0 and x_0 have the same image in V , so the identity gives the square of x_0 , namely $q(c)$. If $P = 1$, then $T = 1$ and the lift of U is a central automorphism of order at most 2, so $g_0 = \sigma_2^2$ acts trivially on the class-two coefficient group and the same identity applies.

The factors u_1^{-1} and x_1^σ have zero V -coordinate on the normalized representative, and hence no quadratic central contribution. Multiplying the table gives (82). Polarizing and using the U -invariance of B gives (84); the operator is invertible by the same polynomial argument used in proposition 5.15. Finally, (85) is exactly (65) on the candidate source. $\square$

6.2. Candidate base Arf invariant.

Lemma 6.6 (Wall doubling). *Let (V, q) be nonsingular over $\mathbf{F}_2$, with polar form B , and let $U \in O(V, q)$ have 2-power order. Put*

$$q_U(x) = q(x) + B(x, U^{-1}x).$$

Then q_U is nonsingular and

$$\text{Arf}(q_U) = \text{Arf}(q) + \text{rank}(1 + U) \pmod{2}. \quad (86)$$

Proof. Let $N = 1 + U$, $R = NV$, and $K = \ker N$. Since $q(Ux) = q(x)$,

$$B(x, U^{-1}x) = B(x, Ux) = q(Nx),$$

so $q_U(x) = q(x) + q(Nx)$. Define the nondegenerate Wall form on R by

$$\omega(Nx, u) = B(x, u).$$

In $(V, q_U) \perp (V, q)$, the diagonal $\{(k, k) : k \in K\}$ is totally singular. Its orthogonal quotient is $R \oplus R$ with quadratic form

$$\Phi_\omega(v, u) = \omega(v, v) + \omega(u, u) + \omega(v, u).$$

If $v \mapsto \omega(v, v)$ is nonzero, split an anisotropic plane from Φ_ω ; if ω is alternating, split a symplectic plane from R , which contributes a hyperbolic four-space. Induction gives $\text{Arf}(\Phi_\omega) = \dim R \pmod{2}$, proving (86). $\square$

Lemma 6.7 (Invariant quadratic forms on a hermitian line). *Let D/D_0 be a quadratic extension of finite fields of characteristic 2, write $x \mapsto x^*$ for its involution, and let $U = \{u \in D^\times : uu^* = 1\}$. Let W be a one-dimensional D -space. Every nonsingular U -invariant quadratic form $Q : W \rightarrow \mathbf{F}_2$ is uniquely of the form*

$$Q(x) = \text{Tr}_{D_0/\mathbf{F}_2}(axx^*) \quad (a \in D_0^\times).$$

Proof. Choose a generator u_0 of the cyclic group U . Since $2^m \equiv -1 \pmod{2^m + 1}$ and no smaller positive power of 2 is 1 modulo $2^m + 1$, the element u_0 has degree $2m$ over $\mathbf{F}_2$; hence $\mathbf{F}_2[u_0] = D$. Invariance gives

$$B(dx, y) = B(x, d^*y) \quad (d \in D).$$

Putting $\ell(y) = B(1, y)$, we obtain $B(x, y) = \ell(x^*y)$. Write $\ell(z) = \text{Tr}_{D/\mathbf{F}_2}(cz)$ for the unique $c \in D$. Symmetry of B forces $c = c^*$, hence $c \in D_0$, and therefore

$$B(x, y) = \text{Tr}_{D_0/\mathbf{F}_2}(c(xy^* + x^*y)).$$

Conversely these forms are alternating, and they are nonsingular exactly when $c \neq 0$. The quadratic form $q_c(x) = \text{Tr}_{D_0/\mathbf{F}_2}(cxx^*)$ has this polarization. Hence $Q - q_c$ is a U -invariant linear form. Writing such a form as $x \mapsto \text{Tr}_{D/\mathbf{F}_2}(bx)$, invariance under u_0 gives $b(u_0 - 1) = 0$, so $b = 0$. Thus $Q = q_c$, uniquely, and nonsingularity gives $c \in D_0^\times$. $\square$

Lemma 6.8 (Ramified Hermitian model and Frobenius fixed space). *Let $I = \langle T \rangle$ be tame inertia, let V be a simple ramified self-dual $\mathbf{F}_2[H_V]$ -module, and write*

$$V|_I \cong W^{\oplus s}, \quad f = \dim_{\mathbf{F}_2} W = 2^a r, \quad r \text{ odd}, \quad a \geq 1.$$

Let q be an H_V -invariant nonsingular quadratic form with polar form B . Then

$$\text{Arf}(q) \equiv s \pmod{2}. \quad (87)$$

For $U = S^{\omega_2}$ one also has

$$\dim_{\mathbf{F}_2} V^U = rs, \quad \text{rank}(1 + U) = rs(2^a - 1) \equiv s \pmod{2}. \quad (88)$$

Consequently the ramified candidate base form of (83) has $\text{Arf}(Q_A^0) = 0$.

Proof. Put

$$D = \text{End}_I(W) \cong \mathbf{F}_{2^f}.$$

Choose an inertia character ζ over $\overline{\mathbf{F}_2}$; then W is the additive group of D and T acts by multiplication by ζ . Since $V|_I$ has only the irreducible isomorphism type W and $V \cong V^\vee$, the inverse character ζ^{-1} lies in the Frobenius orbit of ζ . The character is nontrivial of odd order, so this orbit has even length and

$$\zeta^{-1} = \zeta^{2^{f/2}}.$$

Set

$$D_0 = \mathbf{F}_{2^{f/2}}, \quad x^* = x^{2^{f/2}}.$$

After an I -module identification $V \cong D^s$, the nondegenerate I -invariant form B is uniquely expressible as

$$B(x, y) = \text{Tr}_{D/\mathbf{F}_2}(x^{*t}Hy), \quad H \in \text{GL}_s(D), \quad H^{*t} = H. \quad (89)$$

Indeed, invariance under multiplication by ζ , nondegeneracy of the trace pairing, and symmetry identify the adjoint involution with $*$ and force the matrix to be Hermitian.

Define

$$q_H(x) = \text{Tr}_{D_0/\mathbf{F}_2}(x^{*t}Hx). \quad (90)$$

The scalar $x^{*t}Hx$ lies in D_0 , and

$$\begin{aligned} q_H(x+y) + q_H(x) + q_H(y) &= \text{Tr}_{D_0/\mathbf{F}_2}(x^{*t}Hy + y^{*t}Hx) \\ &= \text{Tr}_{D/\mathbf{F}_2}(x^{*t}Hy) = B(x, y). \end{aligned}$$

Thus $q - q_H$ is linear. It is I -invariant, while $(V^\vee)^I = 0$ because the inertia action is nontrivial and V is simple. Hence $q = q_H$.

Hermitian diagonalization over the finite field extension D/D_0 gives

$$H \sim \text{diag}(a_1, \dots, a_s), \quad a_i \in D_0^\times.$$

Therefore q is an orthogonal sum of the s Hermitian-line forms

$$q_{a_i}(x) = \text{Tr}_{D_0/\mathbf{F}_2}(a_i x x^*).$$

Write $m = f/2$. The norm $D^\times \rightarrow D_0^\times$ is surjective with fibres of size $2^m + 1$, and the trace-zero subspace of D_0 contains $2^{m-1} - 1$ nonzero elements. Hence

$$\#q_{a_i}^{-1}(0) = 1 + (2^m + 1)(2^{m-1} - 1) = 2^{2m-1} - 2^{m-1}.$$

Each Hermitian line has Arf invariant 1. Additivity of the Arf invariant proves (87).

It remains to compute the fixed space of U . Over $\bar{\mathbf{F}}_2$, the f inertia weights form one Frobenius cycle, each weight space having multiplicity s . Let e_ω be the residue modulo $f = 2^a r$ of an integer representing ω_2 . By the defining congruences of ω_2 ,

$$e_\omega \equiv 1 \pmod{2^a}, \quad e_\omega \equiv 0 \pmod{r}.$$

Thus translation by e_ω on $\mathbf{Z}/f\mathbf{Z}$ has r cycles, each of length 2^a . On one cycle write the successive maps as

$$E_0 \xrightarrow{A_0} E_1 \xrightarrow{A_1} \dots \xrightarrow{A_{2^a-1}} E_0, \quad \dim E_i = s.$$

The product $A_{2^a-1} \dots A_0$ is the action of U^{2^a} on E_0 . The shift by $2^a e_\omega$ is trivial modulo f , so U^{2^a} centralizes inertia; it also commutes with S , being a power of S . Hence it is a central 2-element of the faithful tame image. A cyclic 2-group acting in characteristic 2 has nonzero fixed vectors, and centrality makes its fixed space an H_V -submodule. Simplicity therefore forces $U^{2^a} = 1$ on V . Consequently a U -fixed vector on a cycle is uniquely determined by its component in E_0 , and that cycle contributes dimension s . There are r cycles, proving (88).

Finally, lemma 6.6 gives

$$\text{Arf}(Q_A^0) = \text{Arf}(q) + \text{rank}(1 + U) = s + s = 0 \quad \text{in } \mathbf{F}_2.$$

□

Proposition 6.9 (Candidate base determinant zero count). *If $d = \dim V$, then*

$$\#(Q_A^0)^{-1}(0) = \begin{cases} 2^{d-1} - 2^{d/2-1}, & V \text{ unramified,} \\ 2^{d-1} + 2^{d/2-1}, & V \text{ ramified.} \end{cases} \quad (91)$$

Proof. If V is unramified, $U = 1$ and $Q_A^0 = q$. Put $D = \text{End}_{H_V}(V) \cong \mathbf{F}_{2^{2m}}$ and let $D_0 \cong \mathbf{F}_{2^m}$ be the fixed field of the adjoint involution $d \mapsto d^*$ defined by the polar form B . The simple cyclic module V is a one-dimensional D -space. The norm-one group

$$\mathcal{U} = \{u \in D^\times : uu^* = 1\}$$

preserves B . For $u \in \mathcal{U}$, the function $x \mapsto q(ux) + q(x)$ is therefore linear. It is H_V -invariant because u commutes with H_V and q is H_V -invariant. Since V is nontrivial simple, $(V^\vee)^{H_V} = 0$, so the function is zero. Thus $\mathcal{U} \leq O(V, q)$.

By lemma 6.7,

$$q(x) = \text{Tr}_{D_0/\mathbf{F}_2}(axx^*), \quad a \in D_0^\times.$$

The norm fibres have size $2^m + 1$, so the number of zeros is $2^{2m-1} - 2^{m-1}$ and the Arf invariant is 1.

Assume V is ramified. The complete Hermitian and fixed-space calculation is [lemma 6.8](#); in particular $\text{Arf}(Q_A^0) = 0$. The standard zero-count formula for a nonsingular even-dimensional quadratic space gives [\(91\)](#). $\square$

6.3. The local base determinant form and involution-stabilized orbits. Let $q : V \rightarrow \mathbf{F}_2$ be the invariant nonsingular quadratic form under consideration, and let κ_q^0 be the base equivariant determinant class of [lemma 6.3](#). For a local cocycle $b : G_{\mathbf{Q}_2} \rightarrow V$ with lower map ρ , put

$$Q_{\text{loc}}^0([b]) = \text{inv}_{\mathbf{Q}_2}[(b, \rho)^* \kappa_q^0]. \quad (92)$$

This is the base quadratic connecting map. For a general determinant pushout class $\kappa = \kappa_q^0 + \Gamma_{\gamma_\kappa} + \text{inf } \delta_\kappa$, the actual local obstruction is obtained from [\(92\)](#) by the difference formula [\(65\)](#); the affine terms are not folded into the base Gauss sum.

Let H_V be the faithful tame image on V and let $K/\mathbf{Q}_2$ be its tame splitting field. Put $M_K = K^\times/K^{\times 2}$ and let U_i be the image of $1 + \mathfrak{p}_K^i$. If $e = v_K(2)$, then

$$\text{gr}_i M_K \cong \begin{cases} k_K, & 1 \leq i < 2e \text{ odd}, \\ 0, & 1 \leq i < 2e \text{ even}, \end{cases} \quad (93)$$

and

$$U_i^\perp = U_{2e-i+1}, \quad -1 \in U_e. \quad (94)$$

These are the standard dyadic square-class filtration and Hilbert-symbol orthogonality relations; see [\[7, Chapter XIV, Sections 2–3\]](#).

Lemma 6.10 (The middle unit layer is unramified). *Assume $K/\mathbf{Q}_2$ is tame and put $e = v_K(2)$. Then e is odd, and tame inertia acts trivially on the middle graded piece*

$$U_e/U_{e+1}.$$

Consequently a ramified simple tame module does not occur in that graded piece.

Proof. The ramification index of a tame extension of $\mathbf{Q}_2$ is odd and equals the order of tame inertia. On the odd-depth piece $\text{gr}_j M_K$, an inertia element represented by ζ acts through the character $\zeta \mapsto \zeta^j$. At $j = e$ this character is trivial. A ramified simple module has nontrivial inertia, so its isotypic multiplicity at the middle depth is zero. $\square$

Lemma 6.11 (Faithful-image projectivity). *If V is ramified, then V and $V^\vee$ are projective $\mathbf{F}_2[H_V]$ -modules.*

Proof. Let $\overline{\mathbf{F}}_2$ be an algebraic closure, let $I \triangleleft H_V$ be tame inertia, and choose an absolutely irreducible constituent U of $V \otimes_{\mathbf{F}_2} \overline{\mathbf{F}}_2$. Since $|I|$ is odd, $U|_I$ is semisimple. Choose a nontrivial inertia character χ occurring in it, and let H_χ be its stabilizer. Clifford theory expresses U as the induction of a simple H_χ -module on which I acts through χ . The quotient H_χ/I is cyclic. Since χ is H_χ -invariant and $\overline{\mathbf{F}}_2$ contains all roots needed to satisfy the single cyclic relation, χ extends to a one-dimensional character of H_χ . After twisting by that extension, the inducing module factors through the cyclic quotient. Over $\overline{\mathbf{F}}_2$ every simple module for a cyclic group is one-dimensional, and its 2-primary subgroup acts trivially. Thus every χ -weight space in U is one-dimensional.

Let the Frobenius orbit of χ have length $f = 2^a r$ with r odd, and let P be a Sylow 2-subgroup of H_V . The image of P in the cyclic quotient H_V/I is cyclic. Its action on the orbit of χ has 2-power orbit length 2^a . If an element of P stabilizes χ , it acts on the one-dimensional χ -weight space by a scalar of 2-power order, hence trivially because $\overline{\mathbf{F}}_2^\times$ has no nontrivial 2-power torsion. By conjugacy it then acts trivially on all inertia-weight spaces of U , and on every Galois conjugate constituent as well. Faithfulness of the image H_V therefore shows that this stabilizer is trivial. Consequently $|P| = 2^a$, and P acts freely on the weights in blocks of size 2^a .

On each block, transport a nonzero vector around the P -orbit. Because the generator of P has order 2^a , the transported vectors form a regular $\overline{\mathbf{F}}_2[P]$ -basis. Hence the restriction of every absolutely irreducible constituent, and therefore of $V \otimes_{\mathbf{F}_2} \overline{\mathbf{F}}_2$, is a direct sum of regular $\overline{\mathbf{F}}_2[P]$ -modules. It is projective over $\overline{\mathbf{F}}_2[P]$.

Projectivity descends along the faithfully flat extension $\mathbf{F}_2 \subset \overline{\mathbf{F}}_2$, so $V|_P$ is projective over $\mathbf{F}_2[P]$. The Sylow criterion for modular projectivity now gives projectivity over $\mathbf{F}_2[H_V]$. The same argument applied to the inverse inertia characters proves the assertion for $V^\vee$. $\square$

Remark 6.12 (Why the faithful image is required). *Projectivity is asserted only for the faithful image H_V . A larger tame quotient may contain an additional unramified 2-power subgroup acting trivially on V , and projectivity over that larger group can fail. The base determinant connecting map depends only on H_V ; information involving the extra kernel belongs to the inflated and filtration-one components separated from the base determinant calculation.*

Lemma 6.13 (Universal two-point normalization and the index-two Evens norm). *Let*

$$J = \langle s \rangle \cong C_2, \quad E = \mathbf{F}_2 e_1 \oplus \mathbf{F}_2 e_s, \quad s(x_1, x_s) = (x_s, x_1),$$

and put

$$q_{\text{hyp}}(x_1, x_s) = x_1 x_s, \quad f_J(x, y) = x_1 y_s, \quad m_1 = 0, \quad m_s(y) = y_1 y_s.$$

Then

$$\kappa_J((x, c), (y, d)) = f_J(x, cy) + m_c(y) \quad (95)$$

is a normalized 2-cocycle on $E \rtimes J$. It restricts to the quadratic map q_{hyp} on E and to zero on the zero section J . Its fibre extension is the dihedral group D_8 : the two coordinate lifts are involutions, their commutator is the central involution, and their product has square equal to that involution. Moreover

$$[\kappa_J] = N_E^{E \rtimes J, \text{Ev}}(e_1^\vee) \quad \text{in } H^2(E \rtimes J, \mathbf{F}_2). \quad (96)$$

Now let $N \triangleleft G$ have index 2, identify G/N with J , and let $\alpha \in Z^1(N, \mathbf{F}_2)$. Choose lifts $\tilde{1} = 1$ and $\tilde{s} \in G$, and define the normalized Shapiro cocycle

$$b(\gamma)_u = \alpha(\tilde{u}^{-1} \gamma \tilde{\gamma}^{-1} u), \quad u \in J. \quad (97)$$

The graph pullback of (95) is

$$\nu_\alpha(\gamma, \eta) = b(\gamma)_1 b(\eta)_{\tilde{\gamma}^{-1}s} + \varepsilon(\tilde{\gamma}) b(\eta)_1 b(\eta)_s, \quad (98)$$

where $\varepsilon(1) = 0$ and $\varepsilon(s) = 1$. Its class is the index-two Evens norm:

$$[\nu_\alpha] = N_N^{G, \text{Ev}}([\alpha]) \in H^2(G, \mathbf{F}_2). \quad (99)$$

Equivalently, the normalization is

$$N_N^G(1 + [\alpha]) = 1 + \text{cor}_N^G[\alpha] + N_N^{G, \text{Ev}}([\alpha]).$$

Finally, if $g \in H$ is an involution, $M = \mathbf{F}_2[H]$, $J = \langle g \rangle$, and $\pi : M \rightarrow \mathbf{F}_2[J]$ selects the coordinates indexed by 1 and g , then for every right-coset transversal $\mathcal{R}$,

$$[\kappa_g^{\mathcal{R}}] = \text{cor}_{M \rtimes J}^{M \rtimes H}(\pi \rtimes 1)^*[\kappa_J]. \quad (100)$$

Proof. The two identities required in lemma 6.1 are immediate: for $v, w \in E$,

$$m_s(v + w) + m_s(v) + m_s(w) = f_J(sv, sw) + f_J(v, w), \quad m_s(sv) + m_s(v) = 0.$$

Thus (95) is a normalized cocycle. Its square map is $x_1 x_s$, and it vanishes when both fibre coordinates are zero. If $\tilde{e}_1, \tilde{e}_s$ are the coordinate lifts, then $\tilde{e}_1^2 = \tilde{e}_s^2 = 1$, their commutator is the central generator, and $(\tilde{e}_1 \tilde{e}_s)^2$ is that generator; hence the fibre extension is D_8 .

For the swap module,

$$H^1(J, E^\vee) = \frac{\ker(1 + s)}{\text{im}(1 + s)} = 0. \quad (101)$$

The low-degree exact sequence (80) therefore says that a class on $E \rtimes J$ is determined by its restriction to E and its restriction to the zero section. Mackey's formula for the standard Evens norm gives

$$\text{res}_E N_E^{E \rtimes J, \text{Ev}}(e_1^\vee) = e_1^\vee e_s^\vee, \quad \text{res}_J N_E^{E \rtimes J, \text{Ev}}(e_1^\vee) = 0.$$

These are exactly the two restrictions of $[\kappa_J]$, proving (96). Equivalently, this class is the degree-two component of the relative Stiefel–Whitney class $w(\text{Ind } \ell)/w(\text{Ind } 1)$ under the Evens–Kahn normalization; see [9, 10, 11].

For the general index-two inclusion, the graph

$$\Phi_\alpha : G \longrightarrow E \rtimes J, \quad \gamma \longmapsto (b(\gamma), \bar{\gamma})$$

has inverse image of E equal to N , and the pullback of $e_1^\vee$ to N is α . Naturality of the Evens norm in this pullback square and (96) give (99). Direct substitution of (97) into (95) gives (98) term by term. This is also the degree-two term in Evens's normalized bar construction [9, §§4–5]. Changing the lift of s , or changing α by a coboundary, changes both normalized representatives by a normalized coboundary.

For (100), Mackey restriction of the right side to M is

$$\sum_{u \in H/J} x_u x_{ug},$$

and its zero-section restriction is zero. The explicit class $[\kappa_g^{\mathcal{R}}]$ has the same two restrictions. Moreover $H^1(H, M^\vee) = 0$ by Shapiro's lemma, since $M^\vee$ is coinduced. The same low-degree uniqueness argument proves equality. In particular, changing the transversal changes the explicit cocycle only by a normalized coboundary. $\square$

Lemma 6.14 (Regular-module realization of the base local connecting map). *With notation as in lemma 6.3, the base local determinant form satisfies, for every $x \in H^1(\mathbf{Q}_2, V)$,*

$$Q_{\text{loc}, q}^0(x) = Q_{\text{loc}, qW}^0(i_* x). \quad (102)$$

Under Kummer theory and the normalized Shapiro cochain map, the right side is obtained by applying the orbit operations of lemma 6.15 to the scalar Shapiro coordinates. Thus the regular-module computation evaluates the actual base equivariant central class κ_q^0 , including the m_c -terms; it is not merely a polynomial with the same polarization.

Proof. The equality of classes is the pullback identity $\kappa_q^0 = (i \rtimes 1)^* \kappa_{qW}^0$ from (77). Pulling that identity back along a Galois graph cocycle gives equality of normalized central 2-cocycles over $G_{\mathbf{Q}_2}$, hence (102) after applying the local invariant map.

For the second assertion, use the explicit representatives in (75), (76), and (73). Direct substitution of the normalized Shapiro cocycle gives the square and free-orbit corestriction formulas. For a half-orbit, the correction term m_c^g records exactly whether the action on a right coset of $\langle g \rangle$ preserves or reverses the chosen orientation of its two coordinates. Thus the graph pullback is the normalized corestriction of the two-point graph cocycle (98); by lemma 6.13 that two-point class is the index-two Evens norm. The full cochain calculation is written out in lemma 6.15. $\square$

Lemma 6.15 (Quadratic orbit–stabilizer Shapiro). *Let $W = \mathbf{F}_2[H]^N$ have coordinates $X_{r,h}$ with H acting by left translation on h . If K/F is the H -extension and $\alpha_r \in H^1(K, \mathbf{F}_2)$ is the scalar Shapiro coordinate, then the orbit classes of lemma 6.3 evaluate as*

$$S_r \longmapsto \text{cor}_{K/F}(\alpha_r^2), \quad (103)$$

$$C_{r,s,g} \longmapsto \text{cor}_{K/F}(\alpha_r \smile g\alpha_s), \quad (104)$$

$$E_{r,g} \longmapsto \text{cor}_{K_0/F} N_{K/K_0}^{\text{Ev}}(\alpha_r), \quad K_0 = K^{\langle g \rangle}. \quad (105)$$

Proof. Choose lifts $\tilde{h} \in G_F$ of $h \in H$, and write $\bar{\gamma}$ for the image of $\gamma \in G_F$ in H . With the left-regular convention used here, the normalized Shapiro map is

$$b_r(\gamma)_h = \text{Sh}(\alpha_r)(\gamma)_h = \alpha_r(\tilde{h}^{-1} \gamma \widetilde{\bar{\gamma}^{-1} h}).$$

For scalar cocycles α, β on G_K , direct substitution into the normalized bar corestriction gives

$$\sum_{h \in H} \text{Sh}(\alpha)(\gamma)_h \text{Sh}(\beta)(\eta)_{\bar{\gamma}^{-1}h} = \text{cor}_{K/F}(\alpha \smile \beta)(\gamma, \eta). \quad (106)$$

The factor set (75) therefore gives (103). Replacing the second scalar coordinate by $g\alpha_s$ in (106) gives (104).

Now let $J = \langle g \rangle$, let $\mathcal{R}$ be right-coset representatives for H/J , and put $K_0 = K^J$. For $c \in H$ and $u \in \mathcal{R}$, use the notation

$$c^{-1}u = \pi_c(u)g^{\epsilon_c(u)}$$

of lemma 6.2. Let $\rho_0 : G_{K_0} \twoheadrightarrow J$ be the quotient map. The graph pullback of the explicit half-orbit cocycle (73) is

$$\begin{aligned} & \sum_{u \in \mathcal{R}} b_r(\gamma)_u b_r(\eta)_{\bar{\gamma}^{-1}ug} \\ & + \sum_{u \in \mathcal{R}} \epsilon_{\bar{\gamma}}(u) b_r(\eta)_{\pi_{\bar{\gamma}}(u)} b_r(\eta)_{\pi_{\bar{\gamma}}(u)g}. \end{aligned} \quad (107)$$

For completeness, choose the lifts compatibly so that $\widetilde{ug} = \tilde{u} \tilde{g}$ for $u \in \mathcal{R}$, and put

$$\ell_u(\gamma) = \tilde{u}^{-1} \gamma \widetilde{\pi_{\bar{\gamma}}(u)} \in G_{K_0}.$$

Its image in J is $g^{\epsilon_{\bar{\gamma}}(u)}$. In these conventions the normalized corestriction of a 2-cochain ν on G_{K_0} is

$$(\text{cor}_{K_0/F} \nu)(\gamma, \eta) = \sum_{u \in \mathcal{R}} \nu(\ell_u(\gamma), \ell_{\pi_{\bar{\gamma}}(u)}(\eta)). \quad (108)$$

Let $b_J = \text{Sh}_{K/K_0}(\alpha_r)$. Compatibility of the transversals gives

$$b_J(\ell_u(\gamma))_1 = b_r(\gamma)_u$$

and, for $j \in \{1, g\}$,

$$b_J(\ell_{\pi_{\bar{\gamma}}(u)}(\eta))_j = b_r(\eta)_{\pi_{\bar{\gamma}}(u)j}.$$

Substituting the explicit two-point cocycle (98) into (108) gives exactly (107): its first summand is the oriented factor-set term, and its second summand is the orientation-reversal correction. Consequently

$$(b_r, \rho)^* \kappa_g^{\mathcal{R}} = \text{cor}_{K_0/F} \left((\text{Sh}_{K/K_0} \alpha_r, \rho_0)^* \kappa_J \right) \quad (109)$$

as normalized cochains. By lemma 6.13, the inner graph class is $N_{G_K}^{G_{K_0}, \text{Ev}}(\alpha_r)$, which is denoted $N_{K/K_0}^{\text{Ev}}(\alpha_r)$ in field notation. Equation (105) follows. Since every equality above is at the level of normalized cocycles, up to the normalized coboundary caused by a change of transversal, the formulas evaluate the actual orbit classes used in $\kappa_{q_W}^0$, including all m_c -terms. $\square$

Lemma 6.16 (Deep-unit Evens norm). *Let L/k be an unramified quadratic extension of finite dyadic local fields and put $e = v_k(2)$. Then*

$$N_{L/k}^{\text{Ev}}(\alpha) = 0 \quad (\alpha \in U_{e+1}(L) \subset L^\times / L^{\times 2}). \quad (110)$$

Proof. By lemma 6.13, the Evens norm is represented by the explicit two-point graph cocycle (98). The index-two Evens–Kahn formula identifies this normalization with the degree-two term in the relative Stiefel–Whitney identity [10, 11]

$$w(\mathrm{Tr}_{L/k}\langle a \rangle) = w(\mathrm{Tr}_{L/k}\langle 1 \rangle)(1 + \mathrm{cor}_{L/k}[a] + N_{L/k}^{\mathrm{Ev}}([a])). \quad (111)$$

Take a class in $U_{e+1}(L)$ and choose from the outset a representative

$$a = 1 + 2b, \quad b \in \mathfrak{p}_L.$$

Write $L = k(\sqrt{d})$, where d is the unramified quadratic square class, and write

$$a = u + v\sqrt{d}, \quad n = N_{L/k}(a) = u^2 - dv^2.$$

If

$$t = \mathrm{Tr}_{L/k}(b), \quad m = N_{L/k}(b),$$

then

$$u = 1 + t \in \mathcal{O}_k^\times, \quad n = 1 + 2t + 4m. \quad (112)$$

In particular the divisions by u below are legitimate. The trace forms diagonalize as

$$\mathrm{Tr}_{L/k}\langle a \rangle \simeq \left\langle 2u, \frac{2dn}{u} \right\rangle, \quad \mathrm{Tr}_{L/k}\langle 1 \rangle \simeq \langle 2, 2d \rangle.$$

Taking degree two in (111) first gives

$$N_{L/k}^{\mathrm{Ev}}([a]) = (2u, 2dn/u) + (2, 2d) + (d, n).$$

We now keep every Hilbert-symbol cancellation visible. Bilinearity and $(x, x) = (x, -1)$ give

$$\begin{aligned} (2u, 2dn/u) &= (2u, 2) + (2u, d) + (2u, n) + (2u, u), \\ (2u, 2) + (2u, u) &= (2u, 2u) = (2u, -1), \\ (2, 2d) &= (2, -1) + (2, d). \end{aligned}$$

Consequently

$$\begin{aligned} N_{L/k}^{\mathrm{Ev}}([a]) &= (u, -1) + (u, d) + (2u, n) + (d, n) \\ &= (u, -d) + (2du, n). \end{aligned}$$

Thus

$$N_{L/k}^{\mathrm{Ev}}([a]) = (u, -d) + (2du, n). \quad (113)$$

Because L/k is unramified and $b \in \mathfrak{p}_L$, one has $v_k(m) \geq 2$. From (112),

$$\frac{n}{2u-1} = 1 + \frac{4m}{2u-1} \in U_{2e+2}(k). \quad (114)$$

Here $2u-1$ is a unit and $v_k(4m) \geq 2e+2 > 2v_k(2)$. Hensel's lemma, applied to $X^2 - n/(2u-1)$ at $X = 1$, therefore gives $U_{2e+2}(k) \subset k^{\times 2}$. Thus n and $2u-1$ have the same square class.

Every unit of k is a norm from its unramified quadratic extension, and $n = N_{L/k}(a)$. Hence

$$(u, d) = 0, \quad (d, n) = 0.$$

Formula (113) becomes

$$(u, -1) + (2u, 2u-1).$$

The Steinberg relation gives $(2u, 2u-1) = (2u, -1)$. Finally $(2, -1) = 0$: if $i \in k$, then -1 is a square; otherwise $2 = N_{k(i)/k}(1+i)$. Consequently $(2u, -1) = (u, -1)$, and the two terms cancel. $\square$

Lemma 6.17 (The deep half is totally singular). *Assume V is ramified and put*

$$X_+ = \mathrm{Hom}_{H_V}(V^\vee, U_{e+1}) \subset H^1(\mathbf{Q}_2, V).$$

Then

$$\dim X_+ = \frac{1}{2} \dim H^1(\mathbf{Q}_2, V), \quad Q_{\mathrm{loc}}^0|_{X_+} = 0.$$

Proof. By Hochschild–Serre and Kummer theory,

$$H^1(\mathbf{Q}_2, V) \cong \mathrm{Hom}_{H_V}(V^\vee, M_K),$$

and $\mathrm{Hom}_{H_V}(V^\vee, -)$ is exact on the unit filtration. Hilbert duality pairs the depth- j and depth- $(2e - j)$ graded pieces, and self-duality of V gives equal V -isotypic multiplicities in each pair. The only unpaired depth is $j = e$; by [lemma 6.10](#) it has trivial inertia and contains no ramified copy of V . Hence exactly one member of every contributing pair lies above e , proving the dimension statement.

By [lemma 6.14](#), compute the base connecting map after an H_V -split embedding into a regular module, and apply [lemma 6.15](#) to the complete invariant monomial expansion of $q \circ p$. For a class in X_+ every scalar coordinate α_r lies in $U_{e+1}(K)$. Free orbits vanish by [\(94\)](#). Square orbits vanish because $\alpha_r^2 = \alpha_r \smile (-1)$ and $-1 \in U_e$. For an involution orbit, tame inertia has odd order, so K/K_0 in [\(105\)](#) is unramified quadratic. The depth parameter is unchanged in an unramified subextension, and [lemma 6.16](#) makes the Evens norm vanish before corestriction. Every orbit contribution is therefore zero. $\square$

Proposition 6.18 (Dyadic base determinant theorem). *If $d = \dim V$, then*

$$\#(Q_{\mathrm{loc}}^0)^{-1}(0) = \begin{cases} 2^{d-1} - 2^{d/2-1}, & V \text{ unramified,} \\ 2^{d-1} + 2^{d/2-1}, & V \text{ ramified.} \end{cases} \quad (115)$$

Equivalently, the base local Gauss sum is negative in the unramified case and positive in the ramified case. The candidate base form Q_A^0 has the same Gauss sum by [proposition 6.9](#).

Proof. In the ramified case, local Euler–Poincaré and Tate duality give $\dim H^1(\mathbf{Q}_2, V) = d$. By [lemma 6.17](#), the nonsingular base local quadratic space contains a totally singular subspace of dimension $d/2$, so it is hyperbolic.

In the unramified case, the faithful image is cyclic of odd order. Put $D = \mathrm{End}_{H_V}(V)$. Self-duality equips $D \cong \mathbf{F}_{2^{2m}}$ with the involution $x \mapsto x^* = x^{2^m}$ and fixed field $D_0 \cong \mathbf{F}_{2^m}$. Inflation–restriction and the local Euler characteristic identify $H^1(\mathbf{Q}_2, V)$ with a one-dimensional D -space. The norm-one group

$$\mathcal{U} = \{u \in D^\times : uu^* = 1\}$$

preserves the polar form of q . For $u \in \mathcal{U}$, the function $x \mapsto q(ux) + q(x)$ is linear and H_V -invariant. Since $(V^\vee)^{H_V} = 0$, it vanishes; hence every $u \in \mathcal{U}$ is an H_V -equivariant q -isometry. By [\(79\)](#), $(u \rtimes 1)^*[\kappa_q^0] = [\kappa_q^0]$. Functoriality of graph pullback therefore gives

$$Q_{\mathrm{loc}}^0(u_*x) = Q_{\mathrm{loc}}^0(x) \quad (u \in \mathcal{U}).$$

Thus [lemma 6.7](#) applies to the base local form, and it is

$$Q(x) = \mathrm{Tr}_{D_0/\mathbf{F}_2}(axx^*), \quad a \in D_0^\times.$$

The norm map $D^\times \rightarrow D_0^\times$ has fibres of size $2^m + 1$. Among the nonzero elements of D_0 , exactly $2^{m-1} - 1$ have trace 0. Hence

$$\#Q^{-1}(0) = 1 + (2^m + 1)(2^{m-1} - 1) = 2^{2m-1} - 2^{m-1},$$

which is the minus-type number in [\(115\)](#). $\square$

Thus the candidate and local *base* determinant problems have the same Gauss sum. For an arbitrary determinant pushout class, the possible edge γ_κ and inflated scalar class δ_κ are target-side affine data and are handled by the phase-cover argument below.

Corollary 6.19 (Complete source interface). *For every finite lower quotient, every lower exact-image map, and every elementary characteristic-2 coefficient module, the two sources have:*

- (i) *the same obstruction-space dimension and the same number of lifts when the obstruction vanishes;*

- (ii) *natural perfect pairings for all short exact coefficient sequences, with adjoint connecting homomorphisms;*
- (iii) *the same scalar cup–Bockstein form for the trivial module;*
- (iv) *for every nonzero invariant nonsingular quadratic form on a simple self-dual head, the same base determinant Gauss sum for the class κ_q^0 .*

Together with the marked maximal pro-2 identification in [proposition 3.10](#), these are the source-side inputs supplied to the target induction. This corollary is not by itself a determinacy statement: equality of base Gauss sums is used only after Fourier expansion. For a general pushout class

$$\kappa = \kappa_q^0 + \Gamma_{\gamma_\kappa} + \inf \delta_\kappa,$$

the terms γ_κ and δ_κ enter only through the completed square phase classes of [proposition 8.8](#); those phase sums are then converted into smaller central phase covers.

Proof. Items (i)–(iii) are [propositions 5.15](#) and [5.16](#) and [corollary 5.17](#) and (25); item (iv) is [propositions 6.9](#) and [6.18](#). The later operations are restrictions, pushouts, exact-image partitions, finite Fourier transforms, and central-cover constructions internal to the finite target. The determinacy of those operations is isolated in [proposition 8.9](#) and then executed in [section 9](#), with the phase-cover transform (152) supplying the missing affine information. $\square$

6.4. Transgression and an explicit shear.

Lemma 6.20 (No square-only invariant). *Let V be a nontrivial simple $\mathbf{F}_2[C]$ -module. Every nonzero element of $H^0(C, \text{Sym}^2 V^\vee)$ has nonzero, hence nonsingular, polar form.*

Proof. The zero-polar submodule is the Frobenius-square copy of $V^\vee$, whose fixed subspace is zero. The radical of a nonzero invariant polar form is a submodule of the simple module V . $\square$

Lemma 6.21 (Determinant transgression relative to the fixed equivariant class). *Let q be a nonsingular C -invariant quadratic form on V , and assume that a zero-section-normalized equivariant class*

$$\kappa_q^0 \in H^2(V \rtimes C, \mathbf{F}_2)$$

restricting to q on V has been fixed. Let

$$1 \rightarrow V \rightarrow B \rightarrow C \rightarrow 1$$

have extension class $\eta \in H^2(C, V)$. Relative to the fixed equivariant lift κ_q^0 , the obstruction to extending the fibre class over B is

$$d_2(q) = B_{q*}^\flat \eta \in H^2(C, V^\vee). \quad (116)$$

Consequently, if q is the restriction of an actual class in $H^2(B, \mathbf{F}_2)$, then $\eta = 0$ and $B \cong V \rtimes C$ over C .

Proof. Let

$$1 \longrightarrow \mathbf{F}_2 \longrightarrow E_q \longrightarrow V \longrightarrow 1$$

be the central extension represented by $\kappa_q^0|_V$. Central automorphisms of E_q inducing the identity on V are canonically $V^\vee$. If $\hat{v}$ is a lift of $v \in V$, inner conjugation by $\hat{v}$ is the central automorphism

$$w \longmapsto B_q(v, w) = B_q^\flat(v)(w). \quad (117)$$

Choose the zero-section-normalized representative of κ_q^0 . Its preimage of C is split, so conjugation by the chosen lifts of $c \in C$ supplies coherent automorphisms

$$\alpha_c \in \text{Aut}(E_q), \quad \alpha_c \alpha_d = \alpha_{cd},$$

lifting the prescribed C -action on V .

Choose a normalized section $s : C \rightarrow B$ and write

$$s(c)s(d) = \eta(c, d)s(cd). \quad (118)$$

In a putative central extension of B restricting to E_q , lifts of the section elements would satisfy

$$\widehat{s}(c)\widehat{s}(d) = z(c, d)\widehat{\eta(c, d)}\widehat{s}(cd)$$

for a scalar factor $z(c, d) \in \mathbf{F}_2$. Compare the induced automorphisms of E_q . The left side induces $\alpha_c\alpha_d = \alpha_{cd}$, whereas the right side induces

$$\text{Int}(\widehat{\eta(c, d)})\alpha_{cd}.$$

By (117), their discrepancy is the $V^\vee$ -valued normalized 2-cocycle

$$(c, d) \mapsto B_q^b(\eta(c, d)). \quad (119)$$

Changing the section s changes η by a coboundary and changes (119) by the corresponding $V^\vee$ -coboundary. Changing the coherent lifts α_c by central automorphisms $\gamma(c) \in V^\vee$ changes the discrepancy by $d\gamma$. Hence the cohomology class of the obstruction is exactly $B_{q*}^b[\eta]$. This is the $d_2^{0,2}$ differential after the intrinsic equivariance obstruction has been trivialized by the fixed split class κ_q^0 , proving (116).

If an actual class on B restricts to q , the obstruction vanishes. Since q is nonsingular, $B_q^b : V \xrightarrow{\sim} V^\vee$ is an isomorphism of C -modules. Therefore $[\eta] = 0$, so the extension splits over C . $\square$

For a split group $V \rtimes C$, fix the base determinant class κ_q^0 of lemma 6.3. Every central class with restriction q can be written, after changing normalized sections, in the form

$$\kappa = \kappa_q^0 + \Gamma_\gamma + \inf_C^{V \rtimes C} \delta, \quad (120)$$

where $\gamma \in Z^1(C, V^\vee)$, $\delta \in Z^2(C, \mathbf{F}_2)$, and Γ_γ is given by (64).

Lemma 6.22 (Marking-preserving shear of the difference data). *Let $a \in Z^1(C, V)$ and define*

$$s_a(v, c) = (v + a(c), c).$$

If $\kappa = \kappa_q^0 + \Gamma_\gamma + \inf \delta$, then, in cohomology,

$$s_a^*\kappa = \kappa_q^0 + \Gamma_{\gamma+B_q^b a} + \inf(\delta + \Theta_q^0(a) + \gamma \smile a), \quad (121)$$

where

$$\Theta_q^0(a) = (a, \text{id}_C)^*\kappa_q^0 \in Z^2(C, \mathbf{F}_2) \quad (122)$$

and

$$(\gamma \smile a)(c, d) = \gamma(c)(ca(d)). \quad (123)$$

In particular, if $q \neq 0$, a unique cohomology class a kills the edge γ , and the scalar phase then becomes $\delta + \Theta_q^0(a) + \gamma \smile a$.

Proof. The cocycle identity for a is exactly the condition that s_a be an automorphism over C . Substitute $v + a(c)$ and $w + a(d)$ into the normalized representative (120). The terms linear in v or w change the edge by the polar map $B_q^b a$. The term containing only the base class and the graph a is $\Theta_q^0(a)$. The old edge also contributes the scalar term $\gamma(c)(ca(d))$, namely (123). All remaining changes are normalized coboundaries. Since B_q^b is an isomorphism, the edge can be killed uniquely in cohomology.

The shear fixes the quotient C . A marked subgroup containing V is the full inverse image of its image in C , so it is preserved. Scalar decorations kill the nontrivial simple module V , and are therefore preserved as well. $\square$

7. A MINIMAL NONTRIVIAL MODULE LAYER IN THE WILD KERNEL

We now prove the group-theoretic structure needed by the induction. For a finite 2-group K , write

$$\Phi(K) = K^2[K, K]$$

for its Frattini subgroup. A chief factor in the marked kernel will be called *scalar* when it is the trivial $\mathbf{F}_2$ -module. Let G be a framed target with marked normal 2-subgroup L and quotient H . Suppose not every chief factor of L is trivial as an H -module. Choose

$$S \triangleleft P \triangleleft G, \quad P/S \cong V,$$

where all chief factors below S are trivial and V is the first nontrivial simple factor. Choose $K \triangleleft G$, $K \leq P$, minimal subject to $KS = P$, and put

$$R = \Phi(K), \quad M = K/R, \quad T_0 = (K \cap S)/R, \quad C = G/K.$$

Lemma 7.1 (Simple head). *There is an exact sequence of $\mathbf{F}_2[C]$ -modules*

$$0 \rightarrow T_0 \rightarrow M \rightarrow V \rightarrow 0,$$

and

$$T_0 = \text{rad}_{\mathbf{F}_2[C]} M, \quad M/T_0 \cong V, \quad (M^\vee)^C = 0.$$

Proof. Since $K/(K \cap S) \cong V$ is elementary abelian, $R \leq K \cap S$. The displayed sequence follows. Since the quotient is simple, $\text{rad } M \leq T_0$. If a maximal submodule $X < M$ failed to contain T_0 , then $X + T_0 = M$, so X would map onto V . The inverse image of X in K would be a proper G -normal subgroup still satisfying $K_X S = P$, contradicting minimality. Thus every maximal submodule contains T_0 . The assertion about dual invariants follows because a nonzero map $M \rightarrow \mathbf{F}_2$ would give a trivial simple quotient, whereas the unique simple head is V . $\square$

Lemma 7.2 (Frattini–centralizer collapse). *The subgroup R is central elementary abelian in K , and $K^4 = 1$.*

Proof. Let H_2 be the maximal 2-quotient of the finite tame group H , let $N = \ker(H \rightarrow H_2)$, and choose an odd Hall lift $\tilde{N}$ in the preimage of N in G . Along a chief series for S , the action of $\tilde{N}$ on every factor is trivial; coprime action gives

$$[S, \tilde{N}] = 1.$$

Since $R \leq K \cap S$, $\tilde{N}$ centralizes R .

Put $D = C_K(R)$. It is characteristic in K and hence normal in G . The three-subgroup lemma gives $[K, \tilde{N}] \leq D$. The group N acts nontrivially on V : otherwise the action would factor through the 2-group H/N , whose only simple module in characteristic 2 is trivial. Hence $[V, N] = V$, so

$$[K, \tilde{N}]S = P.$$

Thus $DS = P$, and minimality of K gives $D = K$. Therefore $R \leq Z(K)$.

Since K/R is elementary abelian, squares lie in R and commutators have order at most 2. The fourth-power map

$$f : K \rightarrow R, \quad f(k) = k^4,$$

is a G -equivariant homomorphism. The subgroup $\tilde{N}$ fixes R , so $f([k, n]) = 1$ for $k \in K$, $n \in \tilde{N}$. The G -normal closure of $[K, \tilde{N}]$ still maps onto V modulo S , hence equals K by minimality. Thus $f = 1$ and $K^4 = 1$. Finally

$$R = \Phi(K) = K^2[K, K]$$

is central of exponent 2. $\square$

Lemma 7.3 (Decorations vanish on the block). *Every homomorphism from G to an elementary abelian 2-group vanishes on K .*

Proof. It kills $R = K^2[K, K]$ and therefore induces a C -module map $M \rightarrow E$. By lemma 7.1, $\text{Hom}_C(M, \mathbf{F}_2) = 0$, so this map is zero. $\square$

7.1. The determinant pushout on the simple head. Put

$$B = G/R, \quad D_R = (R^\vee)^B = (R^\vee)^C.$$

For $0 \neq \lambda \in D_R$, the pushout $K_\lambda = K/\ker \lambda$ is a central extension

$$1 \rightarrow \mathbf{F}_2 \rightarrow K_\lambda \rightarrow M \rightarrow 1.$$

Let $q_\lambda : M \rightarrow \mathbf{F}_2$ be its square map and b_λ its polar form.

Proposition 7.4 (Simple-head determinant). *For every $0 \neq \lambda \in D_R$,*

$$q_\lambda|_{T_0} = 0, \quad b_\lambda(T_0, M) = 0.$$

Hence q_λ descends to a nonzero nonsingular C -invariant quadratic form

$$\bar{q}_\lambda : V \rightarrow \mathbf{F}_2.$$

Proof. The adjoint $b_\lambda^\flat : M \rightarrow M^\vee$ is C -linear. Since T_0 has only trivial composition factors while $\text{soc}(M^\vee) \cong V^\vee$ is nontrivial simple, $\text{Hom}_C(T_0, M^\vee) = 0$. Thus $b_\lambda(T_0, M) = 0$.

The restriction $q_\lambda|_{T_0}$ is therefore a C -invariant linear functional. Suppose it is nonzero. Quotienting T_0 by its kernel yields a C -module extension

$$0 \rightarrow \mathbf{F}_2 t \rightarrow \bar{M} \rightarrow V \rightarrow 0,$$

with t in the polar radical and $q_\lambda(t) = 1$. Let

$$H_V = \text{im}(C \rightarrow \text{GL}(V)).$$

The kernel of $C \rightarrow H_V$ acts on $\bar{M}$ by possible shears

$$m \mapsto m + f(\bar{m})t.$$

Quadratic invariance and $q_\lambda(t) = 1$ force every such f to vanish, so the extension factors through H_V . If V is ramified, its odd inertia subgroup has no fixed vectors and $H^*(H_V, V^\vee) = 0$ by Hochschild–Serre. If V is unramified, H_V is cyclic of odd order and the same vanishing follows from Maschke’s theorem. Hence

$$H^1(H_V, V^\vee) = 0,$$

so the extension splits. Projection to the scalar summand gives a nonzero C -map $M \rightarrow \mathbf{F}_2$, contradicting $(M^\vee)^C = 0$. Therefore $q_\lambda|_{T_0} = 0$.

The descended form is nonzero. Otherwise K_λ would be elementary abelian, whereas

$$\Phi(K_\lambda) = R/\ker \lambda \neq 1.$$

Its polar radical is a submodule of the simple module V ; if the radical were all of V , the form would be invariant linear, impossible. Thus it is nonsingular. $\square$

8. CENTRAL COVERS, AFFINE FIBRES, AND FOURIER INVERSION

For the simple-head block fixed in section 7, write $T = T_0$ from now on. Thus $0 \rightarrow T \rightarrow M \rightarrow V \rightarrow 0$ is the sequence of lemma 7.1.

Definition 8.1 (Exact-image lower maps). *For a boundary-framed lower target C , let $X_\Gamma(C)$ denote the finite set of boundary-framed exact-image epimorphisms $\rho : \Gamma \twoheadrightarrow C$ at the current lower stage. All sums over $X_\Gamma(C)$ are over exact-image lower maps; proper-image contributions are subtracted separately by induction.*

Lemma 8.2 (Common scalar character group). *For both sources $\Gamma \in \{\Gamma_A, G_{\mathbf{Q}_2}\}$ one has*

$$|\text{Hom}_{\text{cont}}(\Gamma, \mathbf{F}_2)| = 8.$$

Moreover scalar twisting by any such character preserves the boundary-framed condition for the central double covers used below.

Proof. For the candidate, every map to $\mathbf{F}_2$ kills τ because the tame relation forces the tame inertia generator to have odd order in finite quotients; hence it factors through the maximal pro-2 quotient Π . For the local source, the same assertion follows from the characteristic wild subgroup and the common marked maximal pro-2 quotient. The quotient Π has three mod-2 generators, represented by σ, x_0, x_1 , and one Demushkin relator with no linear mod-2 part, so

$$H^1(\Pi, \mathbf{F}_2) \cong \mathbf{F}_2^3.$$

Thus the scalar character group has order $2^3 = 8$ for both sources. The central covers used in the induction have central kernel killed by the fixed boundary and decoration, so multiplying a lift by a scalar character changes only the central coordinate and leaves the boundary-framed data unchanged. $\square$

Lemma 8.3 (Central-cover exact-image transform). *Let $\mathcal{Y} = (Y, L_Y, \pi_Y, \theta_Y)$ be boundary-framed and let*

$$p : \tilde{Y} \twoheadrightarrow Y$$

be a central double cover. Give it the pulled-back structure

$$\tilde{L} = p^{-1}(L_Y), \quad \tilde{\pi} = \pi_Y p, \quad \tilde{\theta} = \theta_Y p,$$

and assume the central kernel lies in $\ker(\tilde{\pi}, \tilde{\theta})$. Fix an exact-image subgroup $J \leq Y$ projecting onto H . If $u_\Gamma^\beta(p, J)$ counts boundary-framed exact-image maps to J whose pullback cover is split, then

$$8u_\Gamma^\beta(p, J) = \sum_{\substack{\tilde{J} \leq p^{-1}(J) \\ p(\tilde{J})=J}} e_\Gamma^\beta(\tilde{J}, \tilde{J} \cap \tilde{L}, \tilde{\pi}|_{\tilde{J}}, \tilde{\theta}|_{\tilde{J}}). \quad (124)$$

Every term on the right is therefore an ordinary exact-image object in the same global boundary-framed category.

Proof. An unobstructed map $f : \Gamma \twoheadrightarrow J$ has a lift, and the set of all lifts is a torsor under $\text{Hom}(\Gamma, \mathbf{F}_2)$. By lemma 8.2, this group has order 8 for both sources, and scalar twisting preserves the boundary-framed condition. Twisting a lift changes only the new central coordinate. Because that coordinate lies in the kernel of both the tame marking and the decoration, every twist still satisfies the identical boundary equation. The exact image of a lift is a subgroup $\tilde{J} \leq p^{-1}(J)$ mapping onto J , and every such exact-image lift occurs in precisely one summand. This is a literal partition of the eight lifts, proving (124); it also proves exact-image and decoration compatibility without an equidistribution assumption. $\square$

Lemma 8.4 (Fourier inversion). *Let D be the character group of a finite $\mathbf{F}_2$ -obstruction space and let $o : X \rightarrow D^\vee$. If*

$$m_\lambda = \#\{x \in X : \langle \lambda, o(x) \rangle = 0\},$$

then

$$\#\{x : o(x) = 0\} = \frac{1}{|D|} \sum_{\lambda \in D} (2m_\lambda - |X|). \quad (125)$$

Proof. Use $1_{o=0} = |D|^{-1} \sum_{\lambda \in D} (-1)^{\langle \lambda, o \rangle}$. $\square$

Lemma 8.5 (Constrained quadratic Gauss transform). *Let W, E be finite $\mathbf{F}_2$ -vector spaces, let $L : W \twoheadrightarrow E$, and let $Q : W \rightarrow \mathbf{F}_2$ be nonsingular with polar form B_Q . For $\kappa \in E$ and $\epsilon \in \mathbf{F}_2$, put*

$$N(\kappa, \epsilon) = \#\{x \in W : Lx = \kappa, Q(x) = \epsilon\}.$$

For $\chi \in E^\vee$, let a_χ be uniquely determined by

$$B_Q(a_\chi, x) = \chi(Lx).$$

Then

$$N(\kappa, \epsilon) = \frac{1}{2|E|} \left(|W| + G(Q) \sum_{\chi \in E^\vee} (-1)^{\chi(\kappa) + \epsilon + Q(a_\chi)} \right), \quad (126)$$

where $G(Q) = \sum_{x \in W} (-1)^{Q(x)}$.

Proof. Expand both conditions with additive characters. Surjectivity of L kills the nontrivial characters in the nonquadratic sum. In the quadratic sum, complete the square:

$$Q(x) + B_Q(a_\chi, x) = Q(x + a_\chi) + Q(a_\chi).$$

This gives (126). $\square$

We now record the two pieces that replace an unrestricted count on all of $H^1(V)$.

Lemma 8.6 (Radical edge, variation formula, and descent). *Let $0 \rightarrow T \rightarrow M \rightarrow V \rightarrow 0$ be the simple-head sequence, and let*

$$p : \tilde{B} \twoheadrightarrow B$$

be a central double cover whose restriction to M has quadratic form with polar radical T and whose restriction to T is zero. The cover determines a canonical edge class

$$[\epsilon] \in H^1(B/T, T^\vee).$$

Its restriction to V is zero, so it is inflated from a unique class

$$[\bar{\epsilon}] \in H^1(C, T^\vee).$$

For every lower exact-image epimorphism $\rho : \Gamma \twoheadrightarrow C$ and every unrestricted M -lift f , twisting by $u \in Z_{\Gamma, \rho}^1(T)$ changes the scalar obstruction by

$$\text{ob}(f_u) = \text{ob}(f) + \langle [u], \rho^*[\bar{\epsilon}] \rangle_{\Gamma}. \quad (127)$$

If $[\bar{\epsilon}] \neq 0$, the free $Z_{\Gamma, \rho}^1(T)$ -action partitions the unrestricted M -lifts into orbits in each of which exactly one half satisfy the central relation. If $[\bar{\epsilon}] = 0$, the cover descends to a central double cover of B/T ; conversely, descent forces the edge class to vanish.

Proof. Let z generate $\ker p$. Since both the square and commutator forms vanish on T , the preimage $\tilde{T} = p^{-1}(T)$ is elementary abelian. Choose a linear complement $s(T)$ to $\langle z \rangle$ in $\tilde{T}$.

For $\bar{b} \in B/T$, choose a lift $\hat{b} \in \tilde{B}$ and define $\epsilon(\bar{b}) \in T^\vee$ by

$$\hat{b} s(t) \hat{b}^{-1} = s(\bar{b}t) z^{\epsilon(\bar{b})(\bar{b}t)}. \quad (128)$$

The definition is independent of the chosen lift $\hat{b}$. Applying (128) successively to two elements shows

$$\epsilon(\bar{b}_1 \bar{b}_2) = \epsilon(\bar{b}_1) + \bar{b}_1 \epsilon(\bar{b}_2),$$

so ϵ is a 1-cocycle. Replacing $s(t)$ by $s(t)z^{\ell(t)}$ changes it by the coboundary $d\ell$; hence its class is canonical.

If $\bar{b} \in V = M/T$, the exponent in (128) is the commutator pairing between a lift of $\bar{b}$ and t . This is the polar form of the restricted quadratic form, and it is zero because T is the polar radical. Hence $\text{res}_V[\epsilon] = 0$. The group $V = M/T$ acts trivially on T , since M is elementary abelian. Inflation–restriction for $1 \rightarrow V \rightarrow B/T \rightarrow C \rightarrow 1$ therefore gives a unique $[\bar{\epsilon}] \in H^1(C, T^\vee)$ inflating to $[\epsilon]$. Choose a normalized cocycle representative $\bar{\epsilon}$ of this class. The difference between ϵ and its inflation is a coboundary $d\ell$ for some $\ell \in T^\vee$; changing the complement $s(T)$ by ℓ therefore lets us assume from now on that $\epsilon = \text{inf}(\bar{\epsilon})$ as cocycles.

Fix a lower exact-image epimorphism $\rho : \Gamma \twoheadrightarrow C$, an M -lift $f : \Gamma \rightarrow B$, and lifts $\hat{f}(g) \in \tilde{B}$. For $u \in Z_{\Gamma, \rho}^1(T)$ put

$$f_u(g) = u(g)f(g), \quad \hat{f}_u(g) = s(u(g))\hat{f}(g).$$

Let ω_f and ω_{f_u} be the scalar central defects. Using (128), the cocycle identity $u(gh) = u(g) + \rho(g)u(h)$, and the fact that $s(T)$ is elementary abelian gives the exact identity

$$\omega_{f_u}(g, h) = \omega_f(g, h) + \bar{\varepsilon}(\rho(g))(\rho(g)u(h)). \quad (129)$$

For the local source, applying the invariant map identifies the second term with the ordinary cup pairing. For the candidate source, its evaluation on the two relation defects is the adjoint pairing supplied by the direct finite-word Fox–Heisenberg chain map of [proposition 5.10](#). Thus (129) yields (127) in either source theory.

If $[\bar{\varepsilon}] \neq 0$, then $\rho^*[\bar{\varepsilon}] \neq 0$. Indeed, a pulled-back coboundary gives the same coboundary formula on every element of C because ρ is surjective. Perfect degree-one duality therefore makes

$$[u] \mapsto \langle [u], \rho^*[\bar{\varepsilon}] \rangle_\Gamma$$

a nonzero linear functional on $H_{\Gamma, \rho}^1(T)$, and hence on $Z_{\Gamma, \rho}^1(T)$. This group acts freely on each fibre of the map from M -lifts to their quotient modulo T . Equation (127) is one affine equation with nonzero linear part on every such orbit, so exactly half of its elements satisfy the scalar central relation.

Finally, if $[\bar{\varepsilon}] = 0$, then $[\varepsilon] = 0$. Alter the complement $s(T)$ so that the cocycle in (128) is identically zero. Then $s(T) \triangleleft \tilde{B}$, and

$$\tilde{B}/s(T) \twoheadrightarrow B/T$$

is a central double cover. Conversely, a descended cover supplies a normal complement to the preimage of T , so the edge class vanishes. $\square$

Assume henceforth that the edge vanishes. By [lemma 6.21](#), the quotient B/T is identified with $V \rtimes C$. For the scalar pushout class under consideration, choose the base determinant class κ_q^0 attached to the descended nonsingular form on V and write the descended central class as

$$\kappa = \kappa_q^0 + \Gamma_{\gamma_\kappa} + \inf_C^{V \rtimes C} \delta_\kappa, \quad \gamma_\kappa \in Z^1(C, V^\vee), \quad \delta_\kappa \in Z^2(C, \mathbf{F}_2). \quad (130)$$

The original group B remains a possibly nonsplit extension of $V \rtimes C$ by T .

Lemma 8.7 (Affine T -lifting equation). *Choose the splitting $B/T \cong V \rtimes C$ above and let*

$$e \in H^2(C, T)$$

be the class obtained by pulling $B \rightarrow V \rtimes C$ back along the zero section $C \rightarrow V \rtimes C$. For a lower exact-image map $\rho : \Gamma \twoheadrightarrow C$, a class

$$c \in H_{\Gamma, \rho}^1(V)$$

is the V -coordinate of an actual M -lift if and only if

$$\partial_{\Gamma, \rho}(c) = \rho^*e \quad \text{in } H_{\Gamma, \rho}^2(T). \quad (131)$$

For every such class, the number of raw lifts above it is

$$\mu = |B_{\Gamma, \rho}^1(V)| |Z_{\Gamma, \rho}^1(T)|, \quad (132)$$

which is independent of ρ and has the same value on the two sources.

Proof. Represent the T -extension of $V \rtimes C$ by a normalized cocycle. Its pure $V \times V$ component is zero because M is elementary abelian; its mixed component is the module extension $0 \rightarrow T \rightarrow M \rightarrow V \rightarrow 0$; and its restriction to the zero section is e . Pulling this cocycle back along the graph of a V -cocycle gives exactly $\partial c + \rho^*e$, proving (131) in characteristic two.

Each cohomology class has $|B^1(V)|$ cocycle representatives, and each solvable T -defect equation has $|Z^1(T)|$ solutions. For the nontrivial simple V , $|B^1(V)| = 2^{\dim V}$; [propositions 5.15](#) and [5.16](#) gives the common value of $|Z^1(T)|$. $\square$

Put

$$D = (T^\vee)^C \cong H_{\Gamma, \rho}^2(T)^\vee.$$

For $\chi \in D$, let

$$\gamma_\chi = \partial_C^\vee(\chi) \in H^1(C, V^\vee), \quad a_{\chi, \kappa} = (B_q^\vee)^{-1}(\gamma_\chi + \gamma_\kappa) \in H^1(C, V).$$

Let

$$\Theta_q^0(a_{\chi, \kappa}) = (a_{\chi, \kappa}, \text{id}_C)^* \kappa_q^0 \in H^2(C, \mathbf{F}_2) \quad (133)$$

be the base graph class, and define the single total target-side phase

$$\Delta_{\chi, \kappa} = \chi_* e + \delta_\kappa + \Theta_q^0(a_{\chi, \kappa}) \in H^2(C, \mathbf{F}_2). \quad (134)$$

Proposition 8.8 (Completed-square phase identity). *For either source, every lower exact-image map ρ , every $b \in H_{\Gamma, \rho}^1(V)$, and every $\chi \in D$,*

$$Q_{\kappa, \Gamma, \rho}(b) + \langle \partial_{\Gamma, \rho} b + \rho^* e, \chi \rangle = Q_{\Gamma, \rho}^0(b + \rho^* a_{\chi, \kappa}) + \iota_\Gamma(\rho^* \Delta_{\chi, \kappa}). \quad (135)$$

Here Q^0 is the base determinant form and Q_κ is the obstruction for the actual scalar pushout class (130).

Proof. By corollary 5.17,

$$\langle \partial_{\Gamma, \rho} b, \chi \rangle = \langle b, \rho^* \gamma_\chi \rangle_\Gamma.$$

Using (65), the left side of (135) is

$$Q_{\Gamma, \rho}^0(b) + \langle b, \rho^*(\gamma_\kappa + \gamma_\chi) \rangle_\Gamma + \iota_\Gamma(\rho^* \delta_\kappa) + \iota_\Gamma(\rho^*(\chi_* e)).$$

By the definition of $a_{\chi, \kappa}$, the middle linear term is the polar pairing $B_q(b, \rho^* a_{\chi, \kappa})$. Completing the square for the base quadratic form gives

$$Q^0(b) + B_q(b, \rho^* a_{\chi, \kappa}) = Q^0(b + \rho^* a_{\chi, \kappa}) + \iota_\Gamma(\rho^* \Theta_q^0(a_{\chi, \kappa})).$$

Adding the two inflated scalar terms proves (135). The class $\Delta_{\chi, \kappa}$ is defined on the finite target C ; no comparison map between the two source H^1 -spaces is involved. $\square$

Proposition 8.9 (Closed exact-image recursion). *Fix a boundary-framed target $\mathcal{Y} = (Y, L_Y, \pi_Y, \theta_Y)$ and abbreviate $e_\Gamma(Y) = e_\Gamma^\beta(\mathcal{Y})$. The same shorthand is used for B , C , and every exact-image subgroup or auxiliary central cover below, always with the boundary frame, marked 2-kernel, tame map, and decoration inherited as in definition 4.1 and lemma 8.3. Suppose first that L_Y has a nontrivial simple head and use the target-theoretic notation*

$$R = \Phi(K), \quad M = K/R, \quad 0 \rightarrow T \rightarrow M \rightarrow V \rightarrow 0, \quad B = Y/R, \quad C = Y/K.$$

The unrestricted M -lifting stage and exact-image subtraction determine $e_\Gamma(B)$ from exact-image counts for strictly smaller targets. If $R = 1$, then $B = Y$ and this is already the desired recursion. Assume henceforth that $R \neq 1$, and put

$$D_R = (R^\vee)^C, \quad z_R = |Z_{\Gamma, \rho}^1(R)| = 2^{2 \dim R + \dim D_R}.$$

The number z_R is independent of the lower exact-image map and has the same value for the candidate and local sources. For $\lambda \in D_R$, let $m_{\Gamma, \lambda}(B)$ be the number of exact-image maps to B for which the scalar pushout $\lambda_ o_R$ vanishes. Then the final R -lifting stage is exactly*

$$e_\Gamma(Y) = \frac{z_R}{|D_R|} \sum_{\lambda \in D_R} (2m_{\Gamma, \lambda}(B) - e_\Gamma(B)). \quad (136)$$

Here $m_{\Gamma, 0}(B) = e_\Gamma(B)$.

For $0 \neq \lambda \in D_R$, let $p_\lambda : B_\lambda \twoheadrightarrow B$ be the scalar central cover and let $Z_{\Gamma, \lambda}(B/C)$ count all compatible lifts of exact-image maps to C , without imposing generation in B . Then

$$m_{\Gamma, \lambda}(B) = Z_{\Gamma, \lambda}(B/C) - \sum_{\substack{J < B \\ J \twoheadrightarrow C}} m_{\Gamma, \lambda}(J). \quad (137)$$

For every proper J in this sum,

$$8m_{\Gamma,\lambda}(J) = \sum_{\substack{\tilde{J} \leq p_\lambda^{-1}(J) \\ p_\lambda(\tilde{J})=J}} e_\Gamma(\tilde{J}). \quad (138)$$

If the radical edge of the scalar cover is nonzero, then

$$Z_{\Gamma,\lambda}(B/C) = 2^{2 \dim M - 1} e_\Gamma(C). \quad (139)$$

If the radical edge is zero, put

$$D = (T^\vee)^C, \quad r = \dim D, \quad d = \dim V, \quad \mu = |B_{\Gamma,\rho}^1(V)| |Z_{\Gamma,\rho}^1(T)|.$$

For the target-side phase classes

$$\Delta_{\chi,\kappa} = \chi_* e + \delta_\kappa + \Theta_q^0(a_{\chi,\kappa}) \in H^2(C, \mathbf{F}_2)$$

one has

$$Z_{\Gamma,\lambda}(B/C) = \frac{\mu}{2^{r+1}} \left(2^d e_\Gamma(C) + G(Q^0) \sum_{\chi \in D} s_\Gamma(\Delta_{\chi,\kappa}) \right), \quad (140)$$

where

$$s_\Gamma(\zeta) = \sum_{\rho \in X_\Gamma(C)} (-1)^{\iota_\Gamma(\rho^* \zeta)}.$$

If $C_\zeta \rightarrow C$ is the scalar phase cover and $n_{\Gamma,0}(\zeta)$ counts the lower exact-image maps for which it is unobstructed, then

$$s_\Gamma(\zeta) = 2n_{\Gamma,0}(\zeta) - e_\Gamma(C) \quad (141)$$

and

$$8n_{\Gamma,0}(\zeta) = \sum_{\substack{\tilde{J} \leq C_\zeta \\ \tilde{J} \rightarrow C}} e_\Gamma(\tilde{J}). \quad (142)$$

Every exact-image count on the right sides of (137)–(142) concerns a target with strictly smaller marked 2-kernel. Therefore the common source interface of [corollary 6.19](#), together with equality for smaller targets, determines $e_\Gamma(Y)$ by the displayed recursion. If all chief factors of L_Y are trivial, the terminal fibre-product argument of [lemma 9.2](#) identifies the two exact-image problems directly.

Proof. The unrestricted elementary M -stage has vanishing obstruction and $2^{2 \dim M}$ lifts over every lower map by [lemma 7.1](#) and [propositions 5.15](#) and [5.16](#). Exact-image subtraction produces the count $e_\Gamma(B)$ and only proper targets satisfying (145). If $R = 1$, then $B = Y$ and this completes the recursion. Assume from now on that $R \neq 1$.

For a fixed exact-image map to B , the set of R -lifts, when nonempty, is a torsor of size z_R . Every such lift is automatically surjective onto Y . Indeed, if its image J maps onto Y/R , then $(J \cap K)R = K$; because $R = \Phi(K)$, the Frattini argument forces $J \cap K = K$, and hence $J = Y$. Thus no further exact-image subtraction is required after the final R -lifting stage. Fourier inversion (125) on the full R -valued obstruction, followed by this common lift multiplicity, gives (136). Partitioning all relation-compatible maps to B by their exact image gives (137), and the literal eight-lift partition of [lemma 8.3](#) gives (138).

When the radical edge is nonzero, the variation formula (127) is a nonzero affine functional on every T -lift torsor, so exactly half of the $2^{2 \dim M}$ unrestricted lifts over each lower exact-image map survive. This is (139).

When the edge is zero, [lemma 6.21](#) identifies B/T with $V \rtimes C$, while [lemma 8.7](#) says that the allowed V -coordinates satisfy $\partial c = \rho^* e$ and occur with multiplicity μ . Applying the constrained Gauss transform (126) and the completed-square identity (135), then summing

over exact lower maps, gives (140). Equation (141) is the definition of the signed phase sum, and (142) is again the exact eight-lift partition of lemma 8.3.

Finally,

$$\begin{aligned} |L_C| &= \frac{|L_Y|}{|K|} < |L_Y|, & |J \cap L_B| &\leq \frac{|L_B|}{2} < |L_Y|, \\ 2|J \cap L_B| &\leq |L_B| < |L_Y|, & |L_{C_\zeta}| &= 2|L_C| = \frac{2|L_Y|}{|K|} < |L_Y|, \end{aligned}$$

the last inequality using $\dim V \geq 2$ and hence $|K| \geq 4$. Thus every recursive call is strictly smaller. The terminal case is lemma 9.2. The formulas therefore form a closed deterministic recursion, rather than a separate unproved assertion that the source interface determines the count. $\square$

9. PROOF OF THE BOUNDARY-FRAMED THEOREM

We prove theorem 4.2 by strong induction on $|L_Y|$, simultaneously for all boundary-framed marked targets. Every recursive target below carries the same map to $H \times E$.

Lemma 9.1 (Coprime-kernel subdirect products). *Let $A \twoheadrightarrow C$ and $B \twoheadrightarrow C$ be finite epimorphisms whose kernels have coprime orders. If a subgroup $J \leq A \times_C B$ projects onto both A and B , then $J = A \times_C B$.*

Proof. The kernel of $J \rightarrow A$ is a subgroup of $\ker(B \rightarrow C)$, and the kernel of $J \rightarrow B$ is a subgroup of $\ker(A \rightarrow C)$. Comparing the two order formulas for J shows that both kernels have their full possible orders, because the two ambient kernel orders are coprime. Thus J contains both kernels and is the full fibre product. $\square$

9.1. The terminal case: only trivial module factors.

Lemma 9.2 (Targets with only trivial module factors). *Suppose every chief factor of L_Y is a trivial H -module. Let H_2 be the maximal 2-quotient of H and let $N = \ker(H \rightarrow H_2)$. Then N has odd order, has a unique normal lift $\tilde{N} \triangleleft Y$ centralizing L_Y , and*

$$Y \cong H \times_{H_2} Q, \quad Q = Y/\tilde{N} \text{ a finite 2-group.} \quad (143)$$

Projection to Q identifies exact-image subgroups of Y projecting onto H with subgroups $Q' \leq Q$ that map onto H_2 .

Proof. The kernel N has odd order by lemma 3.1. Let P be its inverse image in Y . Schur–Zassenhaus gives a complement $\tilde{N} \leq P$. Along a Y -chief series

$$1 = L_0 \triangleleft L_1 \triangleleft \cdots \triangleleft L_m = L_Y,$$

triviality of every factor gives $[L_i, \tilde{N}] \leq L_{i-1}$. Hence $[L_Y, {}_m\tilde{N}] = 1$. Coprime action gives

$$[L_Y, \tilde{N}] = [L_Y, \tilde{N}, \tilde{N}] = \cdots = [L_Y, {}_m\tilde{N}] = 1.$$

Thus $P = L_Y \times \tilde{N}$. Schur–Zassenhaus conjugacy says that all complements are L_Y -conjugate; since L_Y centralizes $\tilde{N}$, the complement is unique and therefore normal in Y .

The quotient $Q = Y/\tilde{N}$ is a 2-group. The two quotient maps define

$$Y \longrightarrow H \times_{H_2} Q.$$

Its kernel lies in both L_Y and $\tilde{N}$, hence is trivial, and the two groups have the same order. This proves (143).

If $J \leq Y$ projects onto H and $Q' = \text{pr}_Q(J)$, then $Q' \twoheadrightarrow H_2$. Inside $H \times_{H_2} Q'$, the subgroup J projects onto both factors. The two kernels have odd and 2-power orders, so lemma 9.1 gives $J = H \times_{H_2} Q'$. The converse is immediate. $\square$

Let $q : Q \rightarrow H_2$ be the quotient map. The tame frame $\alpha : T_{\text{tame}} \rightarrow H$ induces, through the full unramified marking of [proposition 3.14](#), a map

$$\alpha_2 : \Pi \rightarrow H_2.$$

Every boundary decoration kills the odd group $\tilde{N}$ and factors through a map $\bar{\theta} : Q \rightarrow E$.

For either source, the universal property of the pullback gives a bijection between maps $f : \Gamma \rightarrow Y$ with fixed tame frame and maps $g : \Pi \rightarrow Q$ satisfying

$$qg = \alpha_2, \quad \bar{\theta}g = \bar{\psi}. \quad (144)$$

This uses the full $\mathbf{Z}_2$ marking: if $H_2 = C_{2^m}$, the first equation fixes the reduction of the unramified character modulo 2^m , not merely its parity.

The kernels of $H \rightarrow H_2$ and $Q \rightarrow H_2$ have coprime orders. Hence a subgroup of $H \times_{H_2} Q$ that projects onto both factors is the whole pullback. It follows that f is surjective exactly when g is surjective. More generally, projection to Q identifies every exact-image subgroup of Y projecting onto H with a subgroup of Q satisfying (144). Thus the terminal exact-image problems are genuinely identical for the two sources, by the fully marked isomorphism of their maximal pro-2 quotients.

9.2. Lifting through the elementary quotient M . Assume a nonscalar chief factor exists and use the notation of [section 7](#):

$$\begin{aligned} R &= \Phi(K), & M &= K/R, & T &= (K \cap S)/R, & V &= M/T, \\ B &= Y/R, & C &= Y/K, & L_B &= L_Y/R, & L_C &= L_Y/K. \end{aligned}$$

By [lemma 7.3](#), the boundary map kills K , so it descends unchanged to B and C .

For every lower exact-image map $\rho : \Gamma \rightarrow C$, [lemma 7.1](#) and [propositions 5.15](#) and [5.16](#) give

$$H_{\Gamma, \rho}^2(M) = 0, \quad |Z_{\Gamma, \rho}^1(M)| = 2^{2 \dim M}.$$

Thus unrestricted M -lift multiplicities agree. If a proper image $J < B$ maps onto C , then

$$1 \rightarrow J \cap M \rightarrow J \cap L_B \rightarrow L_C \rightarrow 1$$

and

$$|J \cap L_B| = |J \cap M| |L_C| \leq \frac{|M|}{2} |L_C| = \frac{|L_B|}{2} < |L_Y|. \quad (145)$$

Global induction subtracts all proper images and gives equality of the exact-image counts into B .

9.3. Lifting through the Frattini layer R and its scalar central pushouts. Set

$$D_R = (R^\vee)^B = (R^\vee)^C.$$

Lemma 9.3 (Scalar pushouts separate the complete obstruction). *For either source, the dual of the R -valued obstruction space is canonically $D_R = (R^\vee)^C$. If o is an R -valued lifting obstruction, then*

$$o = 0 \iff \lambda_* o = 0 \text{ for every } \lambda \in D_R.$$

For each $\lambda \in D_R$, the corresponding scalar pushout is a central double cover. Thus the characters used in the Fourier inversion below exhaust the full obstruction dual.

Proof. By [lemma 7.2](#), R is elementary abelian and K centralizes R . Hence the Y -action on R factors through $C = Y/K$, and the obstruction lies in the source-specific degree-two group with coefficients in that $\mathbf{F}_2[C]$ -module. Candidate deformation duality and local Tate duality give, respectively,

$$H_{A, \rho}^2(R)^\vee \cong (R^\vee)^C, \quad H_{F, \rho}^2(R)^\vee \cong (R^\vee)^C.$$

For $\lambda \in (R^\vee)^C$, the subgroup $\ker \lambda$ is normal in Y and C acts trivially on $R/\ker \lambda$. Therefore $Y/\ker \lambda \rightarrow Y/R$ is a central double cover, and pairing with λ is exactly its scalar pushout obstruction. Perfectness proves the equivalence. $\square$

For a relation-compatible exact-image map to B , let o_R be its R -valued obstruction. Fix $0 \neq \lambda \in D_R$ and let

$$p_\lambda : B_\lambda = Y / \ker \lambda \twoheadrightarrow B$$

be the scalar central pushout. For $J \leq B$ mapping onto C , let $m_{\Gamma, \lambda}(J)$ count exact-image maps to J with vanishing scalar pushout, and let $Z_{\Gamma, \lambda}(B/C)$ count all compatible lifts of exact-image maps to C , without imposing generation in B . Then

$$m_{\Gamma, \lambda}(B) = Z_{\Gamma, \lambda}(B/C) - \sum_{\substack{J \leq B \\ J \twoheadrightarrow C}} m_{\Gamma, \lambda}(J). \quad (146)$$

For every proper J , [lemma 8.3](#) gives

$$8m_{\Gamma, \lambda}(J) = \sum_{\substack{\tilde{J} \leq p_\lambda^{-1}(J) \\ p_\lambda(\tilde{J}) = J}} e_\Gamma^\beta(\tilde{J}, \dots). \quad (147)$$

Every target on the right has marked kernel at most

$$2|J \cap L_B| \leq |L_B| < |L_Y|, \quad (148)$$

so all proper weighted strata agree by induction.

It remains to compare $Z_{\Gamma, \lambda}(B/C)$. By [proposition 7.4](#), the pushout restricts to a nonsingular determinant form q on the simple head V with scalar radical T .

If the radical edge of [lemma 8.6](#) is nonzero, exactly half of the $2^{2 \dim M}$ unrestricted lifts survive over every lower map. Therefore

$$Z_{\Gamma, \lambda}(B/C) = 2^{2 \dim M - 1} |X_\Gamma(C)|, \quad (149)$$

which agrees by the lower induction.

Suppose the radical edge is zero. Apply [lemmas 6.21](#) and [8.7](#). The quotient B/T is split as $V \rtimes C$, and the scalar pushout class is written relative to the fixed base determinant class as

$$\kappa = \kappa_q^0 + \Gamma_{\gamma_\kappa} + \inf \delta_\kappa.$$

Thus the central obstruction is

$$Q_{\kappa, \Gamma, \rho}(c) = Q_{\Gamma, \rho}^0(c) + \langle c, \rho^* \gamma_\kappa \rangle_\Gamma + \iota_\Gamma(\rho^* \delta_\kappa),$$

but the allowed c are not all of $H_{\Gamma, \rho}^1(V)$: they satisfy the affine equation

$$\partial_{\Gamma, \rho} c = \rho^* e.$$

Put

$$d = \dim V, \quad r = \dim D, \quad D = (T^\vee)^C,$$

and let μ be the common multiplicity in [\(132\)](#). Since $\partial_{\Gamma, \rho}$ is surjective by the exact sequence and $H_{\Gamma, \rho}^2(M) = 0$, [lemma 8.5](#) and [proposition 8.8](#) give, for each lower map,

$$\begin{aligned} & \#\{c : \partial c = \rho^* e, Q_{\kappa, \Gamma, \rho}(c) = 0\} \\ &= \frac{1}{2^{r+1}} \left(2^d + G(Q^0) \sum_{\chi \in D} (-1)^{\iota_\Gamma(\rho^* \Delta_{\chi, \kappa})} \right). \end{aligned} \quad (150)$$

Here $G(Q^0)$ is the common base Gauss sum supplied by [propositions 6.9](#) and [6.18](#); the affine data γ_κ and δ_κ appear only through the single phase class

$$\Delta_{\chi, \kappa} = \chi^* e + \delta_\kappa + \Theta_q^0(a_{\chi, \kappa}).$$

For $\zeta \in H^2(C, \mathbf{F}_2)$ define

$$s_\Gamma(\zeta) = \sum_{\rho \in X_\Gamma(C)} (-1)^{\iota_\Gamma(\rho^* \zeta)}.$$

Summing (150) yields the affine central formula

$$Z_{\Gamma, \lambda}(B/C) = \frac{\mu}{2^{r+1}} \left(2^d |X_{\Gamma}(C)| + G(Q^0) \sum_{\chi \in D} s_{\Gamma}(\Delta_{\chi, \kappa}) \right). \quad (151)$$

This is the unrestricted whole-space formula only when $T = 0$; the scalar phase is counted exactly once in $\Delta_{\chi, \kappa}$.

For any ζ , let $p_{\zeta} : C_{\zeta} \rightarrow C$ be the corresponding central cover, equipped with marked subgroup $p_{\zeta}^{-1}(L_C)$ and with the tame map and decoration pulled back from C . Its central kernel is killed by the boundary map, so C_{ζ} and each exact-image subgroup occurring below are objects of the same boundary-framed category. Let $n_{\Gamma, 0}(\zeta)$ count lower maps for which this cover is unobstructed. Then

$$s_{\Gamma}(\zeta) = 2n_{\Gamma, 0}(\zeta) - |X_{\Gamma}(C)|,$$

and lemma 8.3 gives

$$8n_{\Gamma, 0}(\zeta) = \sum_{\substack{\tilde{J} \leq C_{\zeta} \\ p_{\zeta}(\tilde{J}) = C}} e_{\Gamma}^{\beta}(\tilde{J}, \dots). \quad (152)$$

Every phase cover on the right has marked kernel at most

$$2|L_C| = \frac{2|L_Y|}{|K|} < |L_Y|, \quad (153)$$

because the nontrivial simple head has dimension at least two and hence $|K| \geq 4$.

Lemma 9.4 (Strict decrease of recursive targets). *Every recursive target appearing in the elementary-lift, scalar-cover, and phase-cover steps has marked 2-kernel strictly smaller than the current $|L_Y|$.*

Proof. The lower target has kernel L_Y/K . A proper image in the elementary M -stage satisfies (145), namely $|J \cap L_B| \leq |L_B|/2 < |L_Y|$. A proper scalar-cover stratum satisfies (148), namely $2|J \cap L_B| \leq |L_B| < |L_Y|$. Finally a phase cover satisfies (153). These are exactly the recursive calls used in section 9. $\square$

Thus every phase sum agrees by induction. All factors in (151) agree, so $Z_{\Gamma, \lambda}(B/C)$ agrees. Equation (146) then gives equality of $m_{\Gamma, \lambda}(B)$.

For $\lambda = 0$, the Fourier coefficient is the already matched exact-image count into B . If $R = 1$, the central stage is absent. Fourier inversion (125) gives equality of the complete R -obstruction zero fibre.

Once that obstruction vanishes, both sources have the same R -lift multiplicity. Every such lift is surjective: if its image J maps onto Y/R , then

$$(J \cap K)R = K.$$

Since $R = \Phi(K)$, this forces $J \cap K = K$, and hence $J = Y$.

By lemma 9.4, all recursive calls are to strictly smaller marked 2-kernels. Together with the closed recursion of proposition 8.9, this completes the global induction and proves theorem 4.2.

10. PASSAGE TO ALL FINITE QUOTIENTS

Lemma 10.1 (Exhaustion by tame boundary frames). *Let G be finite and put $L = O_2(G)$. For either source Γ , every epimorphism $f : \Gamma \rightarrow G$ determines a unique tame boundary frame*

$$T_{\text{tame}} \twoheadrightarrow G/L.$$

Conversely, with decoration $E = 0$, a boundary-framed epimorphism to G with a fixed tame frame is exactly an ordinary epimorphism to G inducing that frame. Distinct tame frames give disjoint sets of epimorphisms.

Proof. The characteristic marked wild subgroup maps into $O_2(G) = L$, so every epimorphism has a well-defined quotient map to G/L . By [proposition 3.2](#), this quotient map is a tame frame. Conversely, when $E = 0$ there is no extra decoration, and the boundary-framed condition records only the induced tame quotient map and the marked 2-kernel. Thus a boundary-framed epimorphism with target G is precisely an ordinary epimorphism inducing the chosen frame. The induced quotient map $T_{\text{tame}} \rightarrow G/L$ is unique, so the fixed-frame sets are disjoint as the frame varies. $\square$

Let G be finite and put $L = O_2(G)$. By [lemma 10.1](#), the ordinary surjection set is the disjoint union of the fixed-frame surjection sets over all tame frames $T_{\text{tame}} \rightarrow G/L$. Applying [theorem 4.2](#) to each frame with $E = 0$, summing over the finite set of frames, and using [proposition 2.3](#), gives

$$|\text{Sur}(\Gamma_A, G)| = |\text{Sur}(G_{\mathbf{Q}_2}, G)| \quad (154)$$

for every finite group G .

The candidate is topologically generated by four elements. The one-sided reconstruction lemma [lemma 2.5](#) applied to (154) gives $\Gamma_A \cong G_{\mathbf{Q}_2}$, proving [theorem 1.2](#).

11. COMPLETION AND INDEPENDENCE FROM THE COMPUTATIONAL VERIFICATION

The marked local normalization is proved in [Section 3.1](#); no separate local insertion remains. Its proof uses Labute’s abstract classification, local reciprocity, the intrinsic Demushkin orientation, and the peripheral cyclotomic lifting of [lemmas 3.6](#) and [3.7](#) and [proposition 3.8](#). Consequently the maximal pro-2 quotient of the candidate and $G_{\mathbf{Q}_2}(2)$ agree as marked pairs over the full quotient $\mathbf{Z}_2$.

The finite-word duality, local determinant calculation, constrained Fourier–Gauss transform, strict exact-image induction, and profinite reconstruction are independent of the separate computational verification of the final word. The full integral marking is used only in the construction of the common boundary (26) and in the terminal pro-2 comparison, and those inputs are now supplied by [proposition 1.1](#) and [corollary 3.12](#). Thus the mathematical proof of [theorem 1.2](#) does not depend on the unreproduced computation mentioned in [remark 1.3](#).

APPENDIX A. FINITE FOX–HEISENBERG RULES AND WORD LEDGER

For completeness, this appendix records the finite rules used in [lemmas 5.4](#), [5.5](#) and [5.14](#) and [proposition 6.5](#). All identities here are identities of finite word evaluations before passing to cohomology. Thus there is no appeal to a formal Fox calculus over an infinite completed group. For each finite coefficient calculation one may replace ω_2 by any ordinary integer representative modulo the exponent of the finite semidirect, Heisenberg, or extraspecial group being evaluated; [lemma 5.1](#) then shows that the resulting first and mixed second derivatives are independent of the representative.

Let G act on A and $A^\vee$, and evaluate a word in

$$G \ltimes (A \times A^\vee \times \mathbf{F}_2).$$

If $\bar{u}$ is its base value, $D_u, D_u^\vee$ are its two first derivatives, and β_u is its mixed central coordinate, then direct multiplication in the finite Heisenberg group gives

$$D_{uv} = D_u + \bar{u}D_v, \quad D_{u^{-1}} = -\bar{u}^{-1}D_u, \quad (155)$$

$$D_{u^g} = \bar{g}^{-1}D_u + \bar{g}^{-1}(\bar{u} - 1)D_g, \quad D_{u^n} = \sum_{j=0}^{n-1} \bar{u}^j D_u, \\ \beta_{uv} = \beta_u + \beta_v + (D_u^\vee)(\bar{u}D_v), \quad \beta_{u^{-1}} = \beta_u + (\bar{u}^{-1}D_u^\vee)(\bar{u}^{-1}D_u). \quad (156)$$

The formulas for conjugates, commutators, and powers follow recursively. When $\bar{u}$ has odd order e and the exponent is the 2-primary idempotent, the power sum is the norm projector

$$P = 1 + \bar{u} + \cdots + \bar{u}^{e-1}.$$

We now spell out the first-order ledger at a tame lower map. Write S, T for the lower actions of σ, τ , put $U = S^{\omega_2}$, and let (a, b, c, d) be the lift variables on (σ, τ, x_0, x_1) . The ω_2 -power rule gives

$$D(u_i) = P(Dx_i + D\tau), \quad (157)$$

and hence

$$D(d_0) = Pb + (P + 1)c, \quad D(z_0) = U^{-1}c. \quad (158)$$

The tame relator gives the unnormalized row

$$D(\tau^\sigma \tau^{-2}) = S^{-1}(1 + T)a + (S^{-1} + 1 + T)b. \quad (159)$$

The wild relator has the following first-order table:

factor	a	b	c	d
h_0	0	0	0	0
u_1^{-1}	0	P	0	P
x_1^σ	0	0	0	S^{-1}
$[d_0, z_0]$	0	0	0	0

Thus

$$L_w = Pb + (P + S^{-1})d. \quad (160)$$

Together, (159) and (160) are the Jacobian displayed in [lemma 5.5](#).

The only non-obvious entry in this ledger is the first line. The variation of $g_0 = \sigma_2^2$ contributes only in order at least two, because the lower values of x_0 and d_0 at the tame map are both 1. If $P = 0$, then $Dd_0 = Dx_0 = c$, and the exact class-two identity of [lemma 5.2](#) shows that h_0 has the same elementary and central shadow as x_0^2 . If $P = 1$, then $T = 1$ and the 2-primary operator U acts trivially on the simple characteristic-2 quotient; its lift to the Heisenberg or extraspecial coefficient group is a central automorphism of order at most 2, so $g_0 = \sigma_2^2$ acts trivially on the full coefficient group. Again [lemma 5.2](#) applies. This is the precise reason that h_0 may be replaced by x_0^2 in the linear, mixed cup, and determinant calculations.

For the mixed Heisenberg and extraspecial calculations on normalized representatives supported on x_0 , the ledger is

factor	V -coordinate	central coordinate
h_0	0	$\lambda(c)$ in $H(A)$, or $q(c)$ for κ_q^0
u_1^{-1}	0	0
x_1^σ	0	0
$d_0 = (x_0\tau)^{\omega_2}x_0^{-1}$	$(P + 1)c$	*
$z_0 = x_0^{\sigma_2}$	$U^{-1}c$	*
$[d_0, z_0]$	0	$B((P + 1)c, U^{-1}c)$

The starred central coordinates of d_0 and z_0 are immaterial because these words occur only inside the commutator. The table gives both the mixed Hessian in [lemma 5.14](#) and the exact base determinant word expansion

$$Q_A^0(c) = q(c) + B((P+1)c, U^{-1}c)$$

of [proposition 6.5](#). The starred entries may contain the non-diagonal action corrections m_c from [lemma 6.1](#); they are central coordinates of d_0 and z_0 and hence do not affect the commutator. No relator-by-relator Stokes identity is used: the correction terms in the finite-word Stokes formula cancel only after summing the tame and wild relators, exactly as in [proposition 5.8](#).

APPENDIX B. MACHINE-READABLE SERIALIZATION

For finite computations in groups whose exponents divide

$$85667662080,$$

the idempotent ω_2 is represented by 40491355905. The corresponding machine-readable block is

```
sigma,tau,x0,x1

sigma2 = sigma^40491355905
g0 = sigma2^2
u0 = (x0*tau)^40491355905
u1 = (x1*tau)^40491355905
d0 = u0*x0^-1
z0 = x0^sigma2
c0 = d0^-1*z0^-1*d0*z0
dg = d0^g0
hcomm = dg^-1*d0^-1*dg*d0
h0 = (x0^g0)*x0*dg*d0*d0^2*hcomm

h0*u1^-1*(x1^sigma)*c0
```

The tame relation $\tau^\sigma = \tau^2$ is implicit and is not repeated in the relations section.

The preceding block records the exact word used in the computational verification. The computation is not used in the proof of the marked normalization or any other mathematical step.

APPENDIX C. QUOTIENT AND RECURSION LEDGER

The quotient levels in the nonlinear stage are deliberately distinct:

$$\begin{aligned} R = \Phi(K) \leq K \leq L \leq Y, \quad B = Y/R, \quad C = Y/K, \\ H = Y/L, \quad H_V = \text{im}(C \rightarrow \text{GL}(V)). \end{aligned}$$

Their roles are:

Operation	Quotient retained
Common boundary frame	H and the marked Π through ∂_{bd}
Elementary M -lifting and affine equation	C
Radical edge, transgression, shear, phase classes	C
R -valued obstruction and scalar Fourier pushout	B
Pure determinant zero count	faithful image H_V
Phase-cover transform	central covers $C_\zeta \rightarrow C$

The full quotient C remains visible until every inflated, filtration-one, and affine-lifting component has been encoded. Only the pure determinant zero count passes to H_V . Every auxiliary central cover appearing in the proof has marked-kernel order strictly below the current $|L_Y|$.

APPENDIX D. PROOF DEPENDENCY CERTIFICATE

The following table records the load-bearing implication chain used in the proof. Each row names a proved input and its first downstream use; no computational verification is used in any row.

Proved input	First downstream use
Finite-word Stokes identities, lemma 5.7 and proposition 5.8	Fox–Heisenberg chain map, proposition 5.10
Exact cone dévissage and simple normal forms, lemmas 5.11 and 5.13	duality for arbitrary elementary modules, proposition 5.15
Universal two-point D_8 class, lemma 6.13	normalization of every half-orbit Evens class
Normalized Shapiro–corestriction identity, lemma 6.15	vanishing of the local base form on the deep half, lemma 6.17
Ramified Hermitian and fixed-space calculation, lemma 6.8	candidate ramified Gauss sign, proposition 6.9
Deep-unit Hilbert-symbol ledger, lemma 6.16	local ramified hyperbolicity, proposition 6.18
Qualified determinant transgression, lemma 6.21	splitting $B/T \cong V \rtimes C$ before the affine analysis
Radical-edge variation formula, lemma 8.6	exact half-torsor count in the nonzero-edge case
Closed recursion (136)–(142), proposition 8.9	boundary-framed exact-image theorem, theorem 4.2

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